

Research Collab GROUP35

Members: Renkie Nyeleti Mkhize, Tinashe Nganadange, Gabriel Maponya, Bongumusa Makhubu, Nhlamulo Mabunda , Hluma Nziweni

Please Note: For marking purposes, anyone can be the admin

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ResearchCollab

Project Description

ResearchCollab is a web-based application designed to streamline the end-to-end research lifecycle within universities. It empowers **researchers**, **reviewers**, and **administrators** to collaborate efficiently, manage projects, and track progress with ease. Researchers often face challenges in finding collaborators, managing shared resources, and monitoring progress—ResearchCollab addresses these by centralizing key academic workflows.

Key Features

- **User Verification**
Integration with a third-party identity provider (Google Authentication) limited to @wits.ac.za and @students.wits.ac.za emails. Supports three roles: Researcher, Reviewer, Admin.
- **Project Management**
Create, edit, and delete research projects. Define goals, required skills, timelines, and funding. Invite and manage collaborators, with full control over roles and invitations.
- **Collaboration Tools**
Internal messaging system, proposal invitations, and role-based communication channels. Built-in notifications and file sharing included.
- **Proposal Submission & Review**
Admins can view submitted proposals and assign reviewers. Reviewers submit structured feedback (Approve, Revise, Reject). Proposal status is tracked at every stage.
- **Milestone & Progress Tracking**
Researchers can define milestones, assign responsibilities, and track progress using visual indicators like pie charts.

- **Grant & Funding Tracking**
Create project-specific funding profiles with expense tracking by category. Visual summaries of awarded, spent, and remaining funds.
 - **Report Exporting**
Export dashboards (Projects, Funding, Milestones) as PDFs. Share progress with stakeholders and sponsors.
 - **Customizable Researcher Dashboards**
Add/remove widgets for Projects, Funding, Milestones. Maintain full functionality while personalizing layout.
 - **AI-powered Collaborator Recommendations (Bonus Feature)**
Suggests potential collaborators based on research area, skills, and institutional data.
-

Software Requirements

What Matters? Key Considerations for the Platform

Features

- **User Roles:** Researcher, Reviewer, Admin
- **Project Listings & Invitations**
- **Internal Messaging System**
- **Proposal Review Process**
- **Funding & Milestone Tracking**
- **Exportable Reports & Dashboards**

Security

- **Authentication:**
Uses Google Authentication limited to Wits domain emails.
- **Role-based Access Control:**
Platform permissions differ by user role (e.g., Researchers can't assign reviewers).

Data Sovereignty

- **South African Data Compliance:**
POPIA-compliant, stored locally or in compliant cloud-based solutions.
- **Institutional Ownership:**
Wits retains all user and research data ownership.

Availability & Reliability

- **Development Stage:**
Early-stage deployment with basic login, sign-up, and core features.
- **Hosting:**
Hosted in a lightweight environment to ensure accessibility.
- **Future Scalability:**
CI/CD, monitoring, and uptime guarantees will be implemented in future releases.

User Personas

All users authenticate via Google using Wits-specific emails. Upon signup, users select a role and provide role-specific profile data.

Researcher

Needs:

- Sign up & log in
- Access to project tools and collaboration features

Required on Signup:

- Name
- Email
- Contact Number
- Password
- Department
- Current Academic Role
- Current Research Area
- Research Experience

Admin

Needs:

- Sign up & log in
- Access to all user accounts and review reports

Required on Signup:

- Name
- Email
- Contact Number
- Password
- Department
- Current Academic Role

Reviewer**Needs:**

- Sign up & log in
- Access assigned proposals

Required on Signup:

- Name
- Email
- Contact Number
- Password
- Department
- Current Academic Role
- Research Area
- Research Experience

Feature Breakdown by Sprint**Sprint 1: Basic Account Access and Setup**

- User registration with role-based data
- Authentication and dashboard access
- Secure account creation with validation

Sprint 2: Project & Collaboration Tools

1. **Project Creation and Publishing**
2. **Editing and Removing Projects**
3. **Inviting Collaborators & Managing Invitations**
4. **Viewing Received Invitations and Accepting/Declining**

Each project listing includes full customization options and validation workflows. Researchers manage their project visibility and collaboration teams from a central dashboard.

Sprint 3: Admin, Reviewer, & Messaging Tools

1. **Admin User Management** – View all users, modify roles
2. **Proposal Review Workflow** – Admin assigns, Reviewer evaluates
3. **Reviewer Dashboards** – View, track, and submit evaluations
4. **Internal Messaging System** – Role-based direct messages, file sharing, in-app alerts

Sprint 4: Progress Monitoring & Reporting

1. **Funding Tracker** – Budget breakdowns, categories, and charts
2. **Milestone Management** – Timeline assignment and visual progress
3. **PDF Exporting** – Download reports for Funding, Milestones, and Custom Dashboard
4. **Custom Dashboards** – Widget-based personalization with full functionality

Overall Summary

ResearchBridge functions as a centralized research management hub designed for academic environments. It supports:

- End-to-end research collaboration
- Streamlined communication

- Transparent funding and proposal processes

- Personalized user experiences

With its scalable structure and planned enhancements, the platform is positioned to significantly improve research transparency, teamwork, and academic productivity within institutions like the University of the Witwatersrand.

Team Profiles/Meet the Team

Product Owner

Full Names: Tishra Pillay

Description : The Client who has commissioned the design of the web application

Developers :



Full Names: Renkie Nyeleti Mkhize

Role: Scrum Master

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)

Description:

Ms Mkhize brings exceptional leadership and communication skills to the team, fostering a supportive and productive environment. With a strong background in Mathematics, she values precision and pays close attention to detail, ensuring all aspects of the project are thoroughly considered. Her core value of family drives her commitment to creating a respectful and inclusive team culture.



Full Names: Nhlamulo Delight Mabunda

Role: Database Manager

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)

Description:

Mr Mabunda is a dynamic and highly skilled backend specialist with a strong ability to think on his feet. His curiosity drives him to ask critical questions, and his passion for complex systems allows him to navigate challenges with confidence. He thrives under pressure and consistently delivers thoughtful, high-quality contributions to the team.



Full Names: Bongumusa Makhubu

Role: Github Facilitator

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)

Description:

Mr Makhubu is an insightful and intelligent team member whose problem-solving skills and calm demeanour make him an invaluable asset to the group. Always willing to help, he brings clarity and patience to every task, fostering a thoughtful and collaborative environment. His contributions are central to the smooth operation and cohesion of the team.



Full Names: Tinashe Nganadange

Role: Testing

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)

Description:

Mr Nganadange possesses strong coding skills and a sharp problem-solving mindset. His leadership experience enhances his ability to collaborate effectively and contribute meaningfully to the team dynamic. Calm and focused under pressure, he consistently brings reliability and strength to every stage of development.



Full Names: Hluma Nziweni

Role: UI Designer

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)

Description:

Mr Nziweni is a highly creative individual whose talent shines through his thoughtful UI designs and organized approach. He works exceptionally well in team settings, demonstrating strong collaboration and resource-gathering skills. His eagerness to learn and ability to grasp new concepts quickly make him a valuable and adaptable team member.



Full Names: Gabriel Maponya

Role: Google Authentication

Qualifications:

- ✓ BSc Computer Science (University of Witwatersrand)
- ✓ BSc Honours Degree in Computer Science (University of Witwatersrand)

Description:

Mr Maponya is a skilled and intelligent developer with strong coding expertise. He communicates clearly and effectively, making collaboration seamless. Always willing to assist and consistently dependable, he brings enthusiasm and readiness to every task the team undertakes.

Project Execution Details

Tools

- ✓ React: Coding
 - ✓ MongoDB: Database
 - ✓ Cloud: Microsoft Azure
 - ✓ Project Management Software: Taiga
 - ✓ Collaboration: GitHub
 - ✓ Communication: WhatsApp – Daily Updates
 - MS Teams – Online Meetings
 - Computer Labs – In Person Meetings
-

Testing Plan and Strategy

In our testing plan and strategy, we have chosen to implement it using JEST which works hand in hand with react-testing library. We have chosen to test individual components as well as dashboards so we can show different types of testing such as unit-testing, integration testing and end-end testing. Our tests can be located on “FrontEnd/research-nexus/src/__tests__” and “BackEnd/__tests__”.

We currently have a test-coverage of 78.89%, just shy of the 80% mark.


```

50
51   test('renders form fields', () => {
52     renderWithRouter(<CreateProject />);
53
54     expect(screen.getByLabelText(/Project Title/i)).toBeInTheDocument();
55     expect(screen.getByLabelText(/Description/i)).toBeInTheDocument();
56     expect(screen.getByLabelText(/Research Goals/i)).toBeInTheDocument();
57     expect(screen.getByLabelText(/Research Area/i)).toBeInTheDocument();
58     expect(screen.getByLabelText(/Start Date/i)).toBeInTheDocument();
59     expect(screen.getByLabelText(/End Date/i)).toBeInTheDocument();
60   });
61
62   test('shows additional fields when funding/collaborators checked', () => {
63     renderWithRouter(<CreateProject />);
64
65     const fundingCheckbox = screen.getByLabelText(/Funding Available/i);
66     fireEvent.click(fundingCheckbox);
67     expect(screen.getByLabelText(/Funding Amount/i)).toBeInTheDocument();
68
69     const collabCheckbox = screen.getByLabelText(/Seeking Collaborators/i);
70     fireEvent.click(collabCheckbox);
71     expect(screen.getByLabelText(/Collaborator Roles Needed/i)).toBeInTheDocument();
72   });

```

Examples of Integration tests for CollaborationDashboard.tsx:

```

116
117   test('displays a project card with download link', async () => {
118     mockedAxios.get.mockImplementation((url) => {
119       if (url.includes('/creator/')) {
120         return Promise.resolve({
121           data: [
122             {
123               _id: 1,
124               title: 'AI Research',
125               description: 'A project about AI.',
126               research_goals: 'Improve ML models',
127               research_area: 'Artificial Intelligence',
128               start_date: '2024-01-01',
129               end_date: '2024-12-31',
130               funding_available: true,
131               funding_amount: '10000',
132               collaborators_needed: true,
133               collaborator_roles: 'Researcher',
134               institution: 'Tech University',
135               contact_email: 'test@example.com',
136               created_at: '',
137               file: {
138                 data: 'abc',
139                 contentType: 'application/pdf',

```



```

79
80   test('switches to "Find Opportunities" tab when clicked', async () => {
81     mockedAxios.get.mockResolvedValue({ data: [] });
82
83     renderWithProviders(<CollaboratorDashboard />);
84
85     await waitFor(() => expect(screen.getByText(/welcome back/i)).toBeInTheDocument());
86
87     const tabs = screen.getAllByText(/Find Opportunities/i);
88     const sidebarTab = tabs.find(el => el.tagName.toLowerCase() === 'a');
89     expect(sidebarTab).toBeTruthy();
90     fireEvent.click(sidebarTab!);
91
92     expect(await screen.findByText(/Opportunities Found/i)).toBeInTheDocument();
93   });
94
95
96   test('calls logout when logout button is clicked', async () => {
97     const logoutMock = jest.fn();
98     const contextWithMockLogout = {
99       ...mockAuthContext,
100       logout: logoutMock, // ✅ use the real mock
101     };

```

More tests can be found in the `__tests__` folder (paths listed on above). For more in-depth analysis of our testing, it can be found using this path(FrontEnd/research-nexus/coverage) on our GitHub repository.

SPRINT EXECUTIONS

Sprint 1 Overview:

Researcher

Needs:

- Must be able to sign up and create an account
- Must be able to log in to access the platform
- Requires access to collaboration and project management tools

Required on Signup:

- Name
- Email

- Contact Number
 - Password
 - Department
 - Current Academic Role
 - Current Research Area
 - Research Experience (Qualifications & Current Research Focus)
-

Admin

Needs:

- Must be able to sign up and create an account
- Must be able to log in to manage platform users
- Requires access to all accounts and system reports

Required on Signup:

- Name
 - Email
 - Contact Number
 - Password
 - Department
 - Current Academic Role
-

Reviewer

Needs:

- Must be able to sign up and create an account
- Must be able to log in to review projects
- Requires access to relevant research proposals

Required on Signup:

- Name
- Email

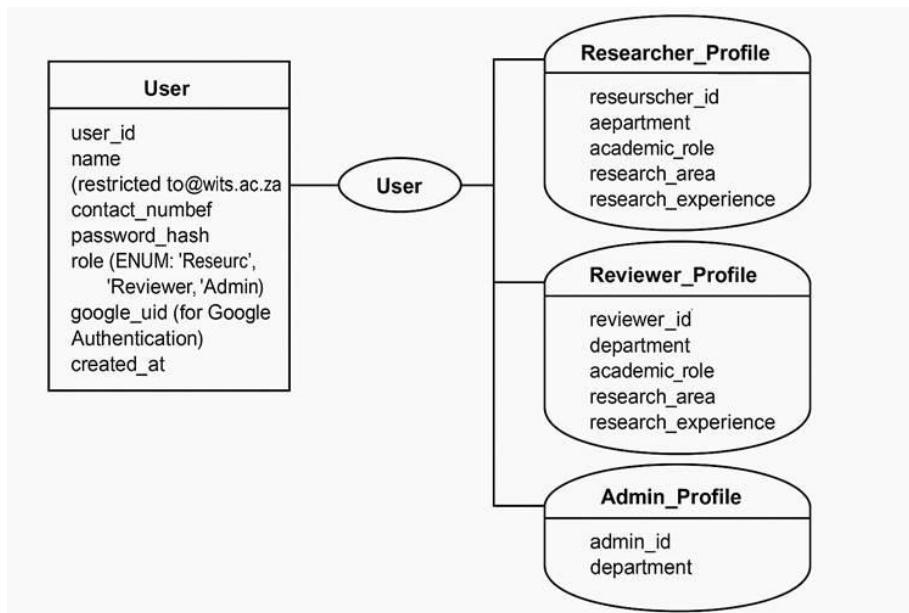
- Contact Number
 - Password
 - Department
 - Current Academic Role
 - Research Area
 - Research Experience
-

UML DIAGRAMS

Mockwire Diagram



Rough idea of a class diagram



User Stories and User Acceptance Test

Original User Story and User Acceptance Tests

User Story 1: Logging In

- **Who:** As a registered user
- **What:** I want to log in
- **Why:** So that I can access the system and use its features

User Acceptance Test:

- **Given** I am a registered user
 - **When** I enter valid login details and click "Login"
 - **Then** I should be taken to the dashboard page
-

User Story 2: Block Unregistered Login

- **Who:** As someone who is not yet registered
- **What:** I want to be blocked from logging in
- **Why:** So that I can first sign up before accessing the system

User Acceptance Test:

- **Given** I am not registered
 - **When** I try to log in
 - **Then** I should see a message telling me to sign up and be redirected to the sign-up page
-

User Story 3: Dashboard After Login

- **Who:** As a logged-in user
- **What:** I want to land on the dashboard
- **Why:** So that I can start working immediately after logging in

User Acceptance Test:

- **Given** I successfully log in
 - **When** the login is complete
 - **Then** I should be taken directly to my dashboard
-

User Story 4: Logging Out

- **Who:** As a logged-in user
- **What:** I want to log out
- **Why:** So that I can end my session and keep my account secure

User Acceptance Test:

- **Given** I am logged in
 - **When** I click "Logout"
 - **Then** I should be logged out and returned to the login page
-

Revised User Stories and User Acceptance test

User Story 1: Researcher Account Registration

As a researcher,
I want to sign up by providing my personal and academic details,
So that I can create an account and access project and collaboration tools.

User Acceptance Test

Given I am a new researcher on the platform,
When I fill out the registration form with my name, email, contact number, password, department, academic role, research area, and experience,
Then my account should be created and I should be redirected to the researcher dashboard.

User Story 2: Admin Account Registration and Access

As an admin,
I want to register and log in with my department credentials,
So that I can manage user roles and oversee platform activity.

User Acceptance Test

Given I have signed up as an admin,
When I log in using valid credentials,
Then I should be able to view all registered users and their roles.

User Story 3: Reviewer Access to Proposals

As a reviewer,
I want to log in and access assigned research proposals,
So that I can review and evaluate them efficiently.

User Acceptance Test

Given I am logged in as a reviewer,
When I navigate to the reviewer dashboard,
Then I should see a list of proposals assigned to me for evaluation.

User Story 4: Researcher Access to Collaboration Tools

As a researcher,
I want to access project creation and team collaboration features after logging in,
So that I can start and manage my research initiatives with other users.

User Acceptance Test

Given I am logged in as a researcher,
When I access the collaboration tools section,

Then I should be able to create a project, invite collaborators, and manage project details.

Meeting Minutes

Original Meeting Minutes

Meeting 1

Meeting Minutes

Project: University Research Collaboration Platform

Meeting Title: Client Requirements Gathering – Meeting 1

Date: 07/04/2025

Time: 14:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Facilitator: Renkie Nyeleti Mkhize

Minute Taker: Hluma Nziweni and Bongumusa Makhubu

Attendees

- Hluma Nziweni
- Gabriel Maponya
- Renkie Nyeleti Mkhize
- Nhlamulo Mabunda
- Tinashe Nganadange
- Bongumusa Makhubu

Client Present: Thishra Pillay

Purpose of Meeting

To gather system requirements and initial expectations from the client for the University Research Collaboration Platform.

Client Requirements Summary

1. Project Management Setup

- Use a project management tool such as **Taiga** or **Notion**.
 - 2. **Git Repository & Commits**
 - Set up a GitHub repository.
 - Each team member must **make at least one commit using their personal email**.
 - 3. **Issue Tracker**
 - Use GitHub Issues to track tasks.
 - Close issues once resolved.
 - 4. **Continuous Integration (CI)**
 - Use **GitHub Actions** or **TravisCI** for CI setup.
 - 5. **Code Coverage Tool**
 - Integrate **Jest** for code coverage tracking.
 - 6. **Unit Testing**
 - **No unit testing required for Sprint 1.**
 - 7. **User Stories**
 - Provide **4 detailed user stories** for initial development.
 - 8. **User Acceptance Tests**
 - Write user acceptance tests based on the user stories.
 - 9. **Project Polish**
 - Prioritize clean, user-friendly design and usability.
 - 10. **Code Documentation**
 - Codebase should be **well documented** to make onboarding new developers easy.
-

Platform Flow Requirements

Landing Page ("Why Download Me")

- Logo
- Basic app description
- Login & Sign-up buttons

Sign-Up Page Requirements

- Direct to a role selection page upon signing up.
- Roles include: **Researcher**, **Reviewer**, and **Admin**.
- Each role has specific required attributes:

Researcher

- Name
- Email
- Contact Number
- Password
- Department
- Academic Role (e.g., lecturer, research coordinator, student)
- Research Area (e.g., AI, Psychology)
- Research Experience (e.g., qualifications, current research)

Admin

- Name
- Email
- Contact Number
- Password
- Department
- Academic Role
- Admin Privilege: manage all accounts (e.g., delete users)

Reviewer

- Name
- Email
- Contact Number
- Password
- Department

- Academic Role
- Research Area
- Research Experience
- **Role Purpose:** Evaluate and give feedback on researcher projects

Signup Notes:

- Include **email verification:** must be a **university email**
- Allow for **third-party verification** (e.g., Google Auth)

Login Page Requirements

- Prompt for **username/email and password**
- On successful login, **redirect to dashboard** (placeholder for now)

Next Steps / Action Items

Task	Due Date
Set up GitHub repository	14/04/2025
Establish CI & Issue Tracker	14/04/2025
Create landing page wireframe	14/04/2025
Draft 4 user stories & UATs	14/04/2025
Design and implement sign-up/login flow	14/04/2025

Next Meeting

Date: 07/04/2025

Time: 15:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Agenda Preview:

1. **Assign Roles Within the Development Team**
 - Define responsibilities (e.g., frontend lead, backend lead, documentation, testing)

2. Draft User Acceptance Test (UAT) Structure

- Finalize format and structure of UATs
- Assign team members to draft initial UATs based on user stories

Meeting 2

Meeting Minutes

Project: ResearchBridge

Meeting Title: Internal Team Meeting – Development Planning

Date: 07/04/2025

Time: 15:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Minute Taker: Hluma Nziweni

Attendees

- Hluma Nziweni
 - Tinashe Nganadange
 - Renkie Nyeleti Mkhize
 - Gabriel Maponya
 - Nhlamulo Mabunda
 - Bongumusa Makhubu
-

Purpose of Meeting

To assign development roles within the group and begin drafting the structure for User Acceptance Tests (UATs).

Discussion Points

1. Role Assignment

- Each team member discussed their strengths and interests.
- Roles were tentatively assigned for more focused collaboration:

- **Google Authentication:** Gabriel Maponya
- **Database Management:** Nhlamulo Mabunda
- **GitHub Management:** Bongumusa Makhuba
- **Testing:** Tinashe Nganadange
- **UI Design:** Hluma Nziweni
- **Documentation:** Renkie Nyeleti Mkhize

2. User Acceptance Test (UAT) Structure

- Agreed to use the **Given-When-Then** format for UATs.
- Discussed aligning UATs directly with user stories.
- Initial UATs will focus on core functionalities:
 - Login
 - Signup
 - Dashboard redirection
 - Logout

3. Next Steps

- Finalize role responsibilities and confirm by next meeting.
- Begin drafting formal UATs and link them to respective user stories.
- Continue work on landing and login pages.

Action Items

Task	Due Date
Confirm and document team roles	13/04/2025
Draft first 4 UATs	13/04/2025
Create basic login/signup functionality	13/04/2025
Design the landing page UI	13/04/2025

Next Meeting

Date: 11/04/2025

Time: 21:30

Location: MSTeams

Agenda Preview:

1. Updates on progress in making the database live
2. GitHub setup and initial commits
3. Progress on Google authentication integration
4. Review of documentation progress
5. UI design updates

Refined Meeting Minutes

DAILY SCRUM MEETINGS

DAILY SCRUM

Meeting Minutes

Project: ResearchBridge

Meeting Title: Internal Team Meeting – Development Planning

Date: 07/04/2025

Time: 15:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Minute Taker: Hluma Nziweni

Attendees

- Hluma Nziweni
- Tinashe Nganadange
- Renkie Nyeleti Mkhize
- Gabriel Maponya

- Nhlamulo Mabunda
 - Bongumusa Makhubu
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 - **Database Management:** Nhlamulo Mabunda
- **GitHub Management:** Bongumusa Makhuba
- **Testing:** Tinashe Nganadange
- **UI Design:** Hluma Nziweni
- **Documentation:** Renkie Nyeleti Mkhize

1. User Acceptance Test (UAT) Structure

- Agreed to use the **Given-When-Then** format for UATs.
- Discussed aligning UATs directly with user stories.
- Initial UATs will focus on core functionalities:
 - Login
 - Signup
 - Dashboard redirection
 - Logout

2. Next Steps

- Finalize role responsibilities and confirm by next meeting.
- Begin drafting formal UATs and link them to respective user stories.

- Continue work on landing and login pages.

Action Items

Task	Due Date
Confirm and document team roles	13/04/2025
Draft first 4 UATs	13/04/2025
Create basic login/signup functionality	13/04/2025
Design the landing page UI	13/04/2025



Next Meeting

Date: 11/04/2025

Time: 21:30

Location: MSTEams

Agenda Preview:

1. Updates on progress in making the database live
 2. GitHub setup and initial commits
 3. Progress on Google authentication integration
 4. Review of documentation progress
 5. UI design updates
-

DAILY SCRUM

Date: 11/04/2025

Time: 21:00

Location: Microsoft Teams

Attendees: Renkie Nyeleti Mkhize, Hluma Nziweni, Nhlamulo Mabunda, Bongumusa Makhubu, Gabriel Maponya, Tinashe Nganadange

Minute Taker: Renkie Nyeleti Mkhize

Facilitator: Renkie Nyeleti Mkhize

Meeting Purpose:

This meeting served to provide project updates and discuss current technical and personal issues impacting development progress.

Individual Updates & Issues:

- **Gabriel Maponya:**
Reported issues with tracking users consistently throughout the app.
 - **Nhlamulo Mabunda:**
Faced challenges in making the database live. Eventually, the team resorted to using Gabriel's database after time constraints.
 - **Bongumusa Makhubu:**
Encountered dependency conflicts due to constant updates to the package.json file.
 - **Tinashe Nganadange:**
Experienced delays starting work due to illness affecting his eyes and overall productivity.
 - **Hluma Nziweni:**
Struggled with coding in React. Described his current skill level as semi-proficient.
 - **Renkie Nyeleti Mkhize:**
Spent time gaining clarity on what documentation fully entails and how to structure it for the project.
-

Action Points:

- Continue to support Gabriel in solving user tracking issues.
- Work towards stabilizing the shared database setup and finalizing its deployment.
- Address package.json dependency alignment.
- Team to provide support and flexibility for Tinashe as he recovers.
- Pair programming or mentorship to assist Hluma with React struggles.
- Renkie to draft and share a documentation outline for feedback.

DAILY SCRUM

Date: 13/04/25

Time: 19:00

Location: Microsoft Teams

Attendees: Renkie Nyeleti Mkhize, Hluma Nziweni, Nhlamulo Mabunda, Bongumusa Makhubu, Gabriel Maponya, Tinashe Nganadange

Minute Taker: Renkie Nyeleti Mkhize

Facilitator: Renkie Nyeleti Mkhize

Discussion Points & Individual Updates:

- **Nhlamulo Mabunda:**
 - Successfully deployed the database online.
 - **Tinashe Nganadange:**
 - Acted as a consultant, helping team members resolve various coding errors.
 - **Bongumusa Makhubu:**
 - Assisted with debugging for Nhlamulo and Gabriel.
 - Performed necessary updates on the GitHub repository.
 - **Hluma Nziweni:**
 - Completed the UI design.
 - Currently facing challenges with linking different pages in React.
 - **Renkie Nyeleti Mkhize:**
 - Completed all assigned tasks except for the ERD diagrams and updating the TAIGA board.
 - Tasked with updating the application's README documentation.
 - **Gabriel Maponya:**
 - Successfully integrated Google Authentication.
-

Next Steps / Action Items:

- Resolve page-linking issues in React (Hluma).
 - Complete ERD diagrams and TAIGA update (Renkie).
 - Finalize README documentation (Renkie).
 - Continue refining the database and debugging as needed.
-

DAILY SCRUM

Meeting Minutes

Date: 13 April 2025

Time: 19:45

Location: Microsoft Teams (MS Teams)

Attendees: Renkie Nyeleti Mkhize, Hluma Nziweni, Nhlamulo Mabunda, Bongumusa Makhuba, Gabriel Maponya, Tinashe Nganadange

Minute Taker: Renkie Nyeleti Mkhize

Facilitator: Renkie Nyeleti Mkhize

Purpose of the Meeting

To provide updates on individual progress, discuss current challenges, brainstorm solutions collaboratively, and support each other in resolving technical issues.

Individual Updates & Issues

- **Gabriel Maponya**
 - Facing deployment test issues.
 - Suggests using mockwires for creating a mock UI.
 - Will integrate Hluma's code into the existing working backend code.
 - Agrees with Musa that debugging and linking components via GitHub workflows is proving difficult.
 - Suggests removing "Login with Apple," "Login with Facebook," and "Forgot Password" features.

- **Nhlamulo Mabunda**
 - No current issues.
 - Suggests creating temporary buttons or UI elements while waiting for frontend to be completed.
 - Offers to assist Hluma by sending him code to tie different parts together.
 - His frontend branch (“Nhlamulo”) will be merged with Gabriel’s backend branch.
- **Bongumusa Makhuba**
 - Issues with package.json and dependency conflicts due to constant updates.
 - Expresses concern about finishing on time due to complexity in linking components.
- **Tinashe Nganadange**
 - No current issues.
 - Suggests working collaboratively by partitioning the work more effectively.
- **Hluma Nziweni**
 - Facing challenges in linking multiple pages (routing issues).
 - Required to complete and push the landing page so Gabriel can begin integration.
 - Agrees with collaborative approach and supports Tinashe's suggestion.
 - Currently using app.jsx to build and preview components live; final code will not use JSX.
- **Renkie Nyeleti Mkhize**
 - Waiting for other components of the project to be completed in order to progress with her part.

Collaborative Discussion & Solutions

- **Tinashe** proposed working collaboratively and partitioning the work to increase efficiency.
- **Hluma** seconded this approach.

- **Nhlamulo** emphasized the benefit of placeholder elements to enable parallel development.
 - **Gabriel** supported the idea with the use of mockwires for front-end development.
 - **Nhlamulo** and **Gabriel** to combine front-end and back-end branches.
 - **Musa** and **Gabriel** shared concerns over time constraints due to debugging and GitHub workflow crashes.
 - Agreement to simplify features (e.g., removing third-party login options) to streamline progress.
-

Action Items:

- Hluma to push current work and finalize the landing page.
 - Nhlamulo to support Hluma with routing code.
 - Gabriel to integrate frontend and backend codebases.
 - Team to simplify features for better progress and efficiency.
 - Members to continue partitioning tasks and supporting each other with ongoing issues.
-

SCRUM PLANNING MEETING

Meeting Minutes

Project: University Research Collaboration Platform

Meeting Title: Client Requirements Gathering – Meeting 1

Date: 07/04/2025

Time: 14:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Facilitator: Renkie Nyeleti Mkhize

Minute Taker: Hluma Nziweni and Bongumusa Makhubu

Attendees

- Hluma Nziweni
- Gabriel Maponya
- Renkie Nyeleti Mkhize
- Nhlamulo Mabunda
- Tinashe Nganadange
- Bongumusa Makhubu

Client Present: Thishra Pillay

Purpose of Meeting

To gather system requirements and initial expectations from the client for the University Research Collaboration Platform.

Client Requirements Summary

11. Project Management Setup

- Use a project management tool such as **Taiga** or **Notion**.

12. Git Repository & Commits

- Set up a GitHub repository.
- Each team member must **make at least one commit using their personal email**.

13. Issue Tracker

- Use GitHub Issues to track tasks.
- Close issues once resolved.

14. Continuous Integration (CI)

- Use **GitHub Actions** or **TravisCI** for CI setup.

15. Code Coverage Tool

- Integrate **Jest** for code coverage tracking.

16. Unit Testing

- **No unit testing required for Sprint 1.**

17. User Stories

- Provide **4 detailed user stories** for initial development.

18. User Acceptance Tests

- Write user acceptance tests based on the user stories.

19. Project Polish

- Prioritize clean, user-friendly design and usability.

20. Code Documentation

- Codebase should be **well documented** to make onboarding new developers easy.
-

Platform Flow Requirements

Landing Page ("Why Download Me")

- Logo
 - Basic app description
 - Login & Sign-up buttons
-

Sign-Up Page Requirements

- Direct to a role selection page upon signing up.
- Roles include: **Researcher**, **Reviewer**, and **Admin**.
- Each role has specific required attributes:

Researcher

- Name
- Email
- Contact Number
- Password
- Department
- Academic Role (e.g., lecturer, research coordinator, student)
- Research Area (e.g., AI, Psychology)

- Research Experience (e.g., qualifications, current research)

Admin

- Name
- Email
- Contact Number
- Password
- Department
- Academic Role
- Admin Privilege: manage all accounts (e.g., delete users)

Reviewer

- Name
- Email
- Contact Number
- Password
- Department
- Academic Role
- Research Area
- Research Experience
- **Role Purpose:** Evaluate and give feedback on researcher projects

Signup Notes:

- Include **email verification:** must be a **university email**
- Allow for **third-party verification** (e.g., Google Auth)

Login Page Requirements

- Prompt for **username/email and password**
- On successful login, **redirect to dashboard** (placeholder for now)

Next Steps / Action Items

Task	Due Date
Set up GitHub repository	14/04/2025
Establish CI & Issue Tracker	14/04/2025
Create landing page wireframe	14/04/2025
Draft 4 user stories & UATs	14/04/2025
Design and implement sign-up/login flow	14/04/2025

Next Meeting

Date: 07/04/2025

Time: 15:30

Location: University of the Witwatersrand, MSL Lab 005 (In-person)

Agenda Preview:

3. Assign Roles Within the Development Team

- Define responsibilities (e.g., frontend lead, backend lead, documentation, testing)

4. Draft User Acceptance Test (UAT) Structure

- Finalize format and structure of UATs
- Assign team members to draft initial UATs based on user stories

BACKLOG REFINEMENT

Backlog Refinement Meeting: 13 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange

• Clarification of User Stories:

- o User story for **Researcher Sign-Up** was clarified to include full details on required fields: Name, Email, Contact Number, Password, Department, Current Academic Role, Current Research Area, and Research Experience.
- o User story for **Admin Platform Access** was updated to include account creation, role management rights, and visibility into platform reports.

- o User story for **Reviewer Proposal Access** was confirmed to include login functionality and viewing assigned research proposals only.
- o User story for **Cross-role Secure Login** was added to ensure proper role-based access and dashboard routing post-login.

• **User Stories:**

1. As a **Researcher**, I want to sign up by providing all my academic and research details, so that I can gain access to collaboration tools and begin publishing projects.
2. As an **Admin**, I want to register and log in securely, so that I can manage user accounts and system reports effectively.
3. As a **Reviewer**, I want to create an account with detailed expertise fields, so that I can be assigned suitable proposals for review.
4. As a **User** (Researcher, Admin, or Reviewer), I want to log in and be redirected to a personalized dashboard based on my role, so that I can carry out my duties without confusion.

• **User Acceptance Tests (Given-When-Then format):**

- **Test 1 – Researcher Sign-Up:**
Given a researcher has completed all required fields on the sign-up form,
When they click the “Create Account” button,
Then their account should be created, and they should be redirected to the researcher dashboard.
- **Test 2 – Admin Role Login Access:**
Given an admin has a valid account,
When they enter their credentials on the login screen,
Then they should be redirected to the admin panel with access to all user and report tools.
- **Test 3 – Reviewer Registration:**
Given a reviewer has filled out the signup form including research area and experience,
When they submit their registration,
Then their profile should be created and made available for admin assignment to proposals.
- **Test 4 – Secure Role-Based Routing Post Login:**
Given a user logs in with valid credentials,
When the system verifies their role,

Then they should be directed to the correct dashboard (Researcher, Admin, or Reviewer) based on their user type.

- **Prioritization:**

- o High priority was given to completing role-based registration and authentication flows for all user types.

- o Medium priority was assigned to improving form validation and error handling for user signup.

- **Preparation for Sprint:**

- o Tasks ready for Sprint: Researcher and Admin Signup Forms, Reviewer registration flow, Role-based login redirection.

- o Deferred: UI enhancements to dashboards based on user feedback from prototype testing.

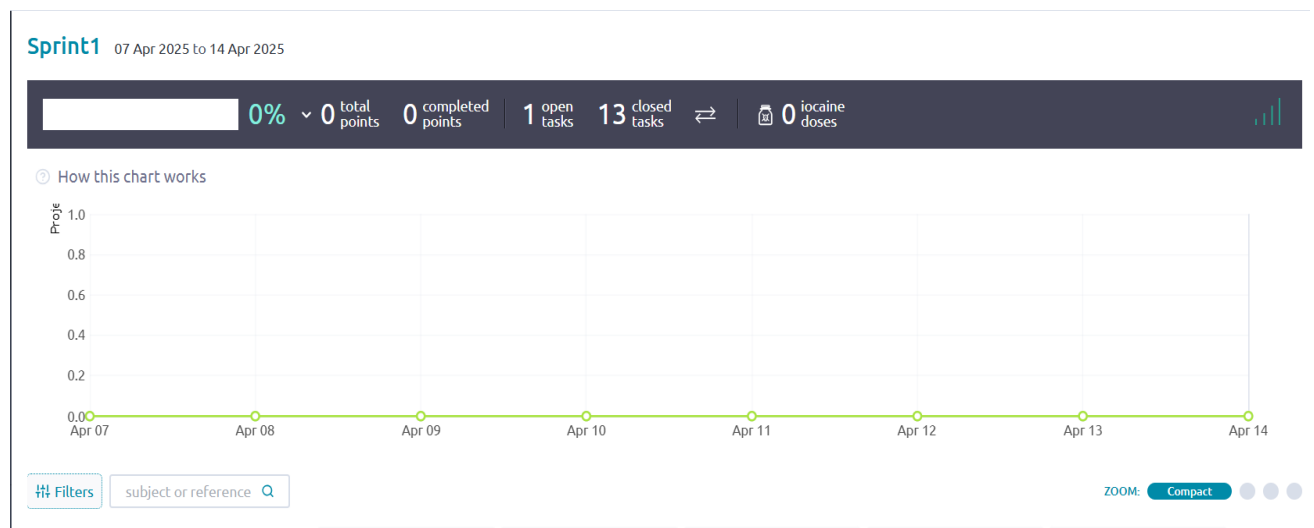
- **New Items Added to Backlog:**

- o Add email verification on account creation.

- o Include tooltips on signup forms for clarity on required fields.

- o Add visual loading indicators on login actions for user feedback.

Sprint burndown chart



Sprint 2 Overview:

Sprint 2: Feature Implementation Summary

Deliverables

- UML Diagrams
 - Unit Tests
 - Six User Stories (detailed below)
-

1. Project Creation and Publishing

As a researcher,

I need the ability to build and share new project listings,

So that I can connect with others for collaborative research opportunities.

Functional Requirements:

a. Project Creation Form

- The platform should provide a user-friendly form for researchers to initiate a new project.
- The form should contain:
 - Project Title (text field)
 - Project Description (text area)
 - Research Goals (text area)
 - Key Research Areas (multi-select dropdown or tags)
 - Start Date (date picker)
 - End Date (date picker)
 - Funding Available (optional checkbox or currency field)
 - Contact Information (email or profile link)
 - Requirements Information:
 - Skills/Expertise Needed (e.g., Data Analysis, Machine Learning)
 - Experience Level (Beginner, Intermediate, Expert)

- Positions Required (e.g., Project Manager, Research Assistant)
- Technical Requirements (e.g., Python, MATLAB)

b. Input Validation

- All required fields must be filled out before submission.
- Provide user feedback for incorrect or missing input.

c. Publishing Workflow

- Once all inputs are validated, the researcher can publish the project.
 - A verification step (e.g., confirmation modal) should be included before final submission.
-

2. Editing Project Details

As a researcher,

I want to update the information on my existing projects,

So that I can keep project listings accurate and current.

Functional Requirements:

- a. From the dashboard, the researcher can access **"My Projects"** and select **"View All"** to see a list of created projects.
 - b. Clicking on a specific project brings up the full details with an **Edit** button.
 - c. The researcher can modify any of the previously entered information with existing validation rules still enforced.
 - d. A **"Save Changes"** button allows them to confirm and update the project.
-

3. Removing Projects

As a researcher,

I would like the option to delete my projects,

So that I can remove those that are no longer relevant or active.

Functional Requirements:

- a. Accessed the same way as editing—via **"My Projects"** → **View All** → click individual project.
- b. A **Delete** button is provided next to Edit.

c. Clicking delete triggers a confirmation message ("Are you sure you want to delete this project?") with a **Cancel** option.

4. Sending Collaboration Invites

As a researcher,

I want to invite other users to work with me on a project,

So that I can build a team with the right skills and experience.

Functional Requirements:

- a. On the individual project page, there is an **"Invite Collaborators"** button.
 - b. Clicking it opens a **search feature** to find users by:
 - Name
 - Skillset
 - Position
 - c. The results display collaborator profiles; the researcher can select one and proceed with sending an invitation.
 - d. The invitation includes:
 - A **text field** to write a custom message
 - A **predefined or custom role** the invitee is being invited to fill
 - e. Sending the invitation should trigger a **notification** for the selected collaborator.
-

5. Monitoring Sent Invitations

As a researcher,

I need a way to view and manage the invitations I've sent,

So that I can track responses and make adjustments if needed.

Functional Requirements:

- a. Within each project listing, a **"Manage Invitations"** section is available.
- b. This section shows all invitations sent with:
 - Invitee Name or Email
 - Role Assigned
 - Status: Pending, Accepted, Declined, Cancelled, Expired

- Date Sent

c. Options to **resend** or **cancel** invitations may be included for improved control.

6. Handling Received Invitations

As a researcher,

I want to view and respond to invitations from others,

So that I can decide whether to participate in their research based on relevance and my availability.

Functional Requirements:

a. The dashboard features a "**View Invitations**" button, leading to a full page of all received invites.

b. Each invitation includes:

- Project Title
- Inviting Researcher's Name
- Role being offered

c. The recipient can choose to **Accept** or **Decline** the invitation.

d. Expired or Cancelled invitations should be clearly indicated in the UI.

USER STORIES AND USER ACCEPTANCE

Original User Stories and User Acceptance

1. View Research Projects

User Story:

As a researcher,

I want to view a list of all my research projects on a dashboard,

so that I can easily track their details and manage my work in one place.

User Acceptance Test:

Given I am a logged-in researcher with existing projects,

When I navigate to the dashboard,

Then I should see a list of my projects displaying titles, descriptions, research areas, and dates.

2. Create New Research Projects

User Story:

As a researcher,

I want to create new research projects by filling in a detailed form,

so that I can formally document my project goals and timelines from the start.

User Acceptance Test:

Given I am on the “Create Project” page,

When I fill in the required fields (title, description, goals, start/end dates, research area, etc.) and submit,

Then the project should be created and appear on my dashboard with all entered details.

3. Specify Collaboration Needs

User Story:

As a project owner,

I want to specify the roles I need collaborators for when creating a project,

so that I can attract the right people with relevant expertise to work with me.

User Acceptance Test:

Given I am creating a new project,

When I indicate that I need collaborators and specify roles (e.g., data analyst, statistician),

Then those roles should be listed under the project and visible to potential collaborators browsing projects.

4. Manage Research Funding Information

User Story:

As a researcher,

I want to indicate if funding is available and specify the amount,

so that potential collaborators understand the financial support available for the project.

User Acceptance Test:

Given I am creating or editing a project,

When I check the option for available funding and input an amount,
Then this information should be saved and shown on the project card for visibility.

5. Track Project Timelines

User Story:

As a researcher,
I want to set start and end dates for my projects,
so that I can monitor progress and stay on schedule.

User Acceptance Test:

Given I am filling out the project creation form,
When I select valid start and end dates,
Then the timeline should be recorded and displayed clearly on the project dashboard and details page.

6. Filter and Categorize Research by Field

User Story:

As a researcher,
I want to categorize my research projects by selecting a research area,
so that my projects are well-organized and easy for others with similar interests to find.

User Acceptance Test:

Given I am creating a new project,
When I select a research area from the dropdown (e.g., AI, Psychology, Bioinformatics),
Then a badge with that category should appear on the project card, and the project should be searchable by that field.

Refined User Stories and User Acceptance Tests

1. Project Creation and Publishing

User Story:

As a researcher,

I need the ability to build and share new project listings,
So that I can connect with others for collaborative research opportunities.

User Acceptance Test:

Given that I am a logged-in researcher

And I am on the “Create Project” page

When I fill in all the required fields and click the “Publish” button

Then the new project should be saved and listed under “My Projects”

And it should become visible to other users (based on visibility settings)

2. Editing Project Details

User Story:

As a researcher,

I want to update the information on my existing projects,

So that I can keep project listings accurate and current.

User Acceptance Test:

Given that I am a logged-in researcher

And I have at least one existing project listed under “My Projects”

When I click on a project and choose “Edit”

And I update one or more fields and click “Save Changes”

Then the updated details should be reflected in the project listing

3. Removing Projects

User Story:

As a researcher,

I would like the option to delete my projects,

So that I can remove those that are no longer relevant or active.

User Acceptance Test:

Given that I am a logged-in researcher

And I have a project listed under “My Projects”

When I click the “Delete” button and confirm the deletion

Then the project should be permanently removed from my list

And it should no longer be visible to others on the platform

4. Sending Collaboration Invites

User Story:

As a researcher,
I want to invite other users to work with me on a project,
So that I can build a team with the right skills and experience.

User Acceptance Test:

Given that I am a logged-in researcher
And I am viewing one of my project pages
When I click “Invite Collaborators”
And I select a user and specify a role and message
Then the selected user should receive a notification with the invitation details

5. Monitoring Sent Invitations

User Story:

As a researcher,
I need a way to view and manage the invitations I’ve sent,
So that I can track responses and make adjustments if needed.

User Acceptance Test:

Given that I am a logged-in researcher
And I have previously sent collaboration invitations for a project
When I visit the “Manage Invitations” section for that project
Then I should see a list of all invitations with their status (Pending, Accepted, Declined, etc.)
And I should be able to resend or cancel an invitation

6. Handling Received Invitations

User Story:

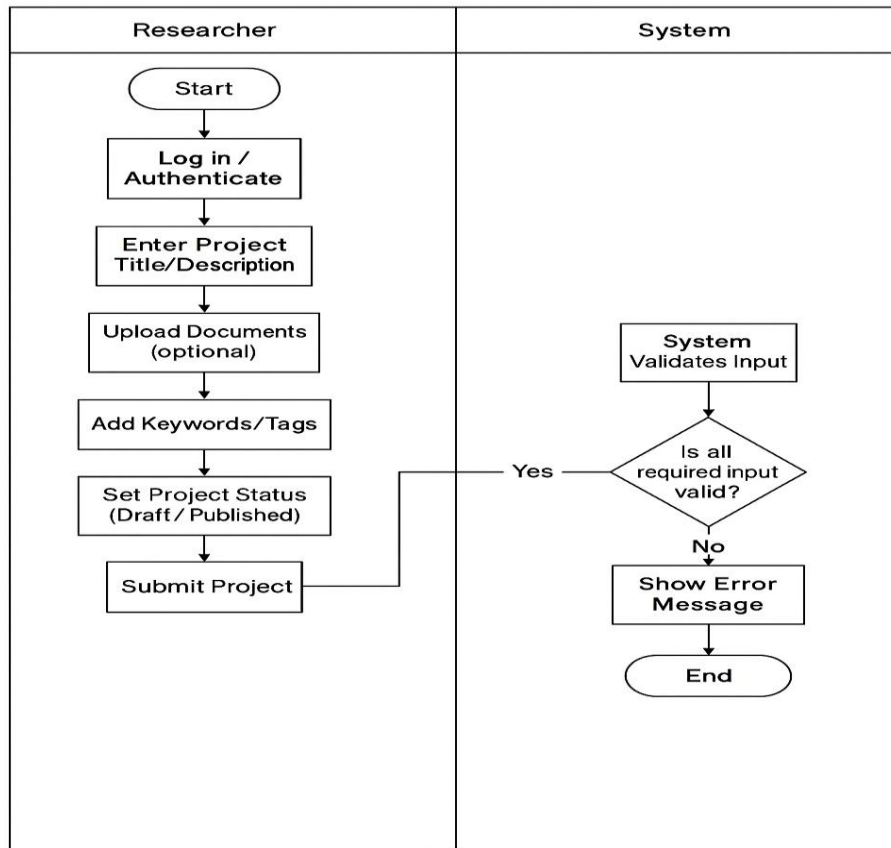
As a researcher,
I want to view and respond to invitations from others,
So that I can decide whether to participate in their research based on relevance and my availability.

User Acceptance Test:

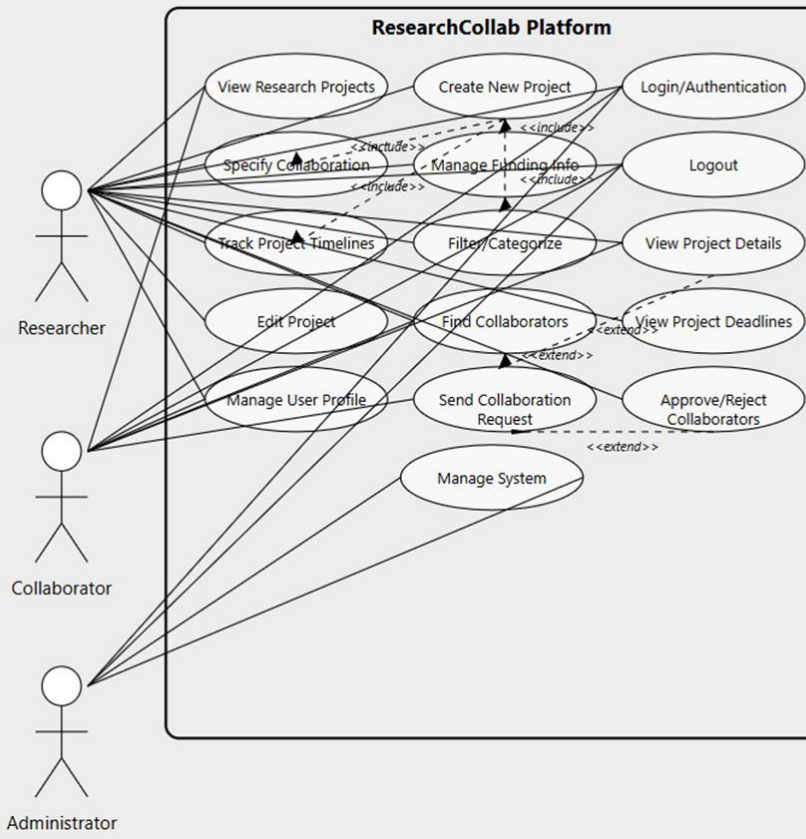
Given that I am a logged-in researcher
And I have received one or more collaboration invitations
When I go to the “View Invitations” section from my dashboard
Then I should see a list of all received invitations with project and role details
And I should be able to accept or decline each invitation

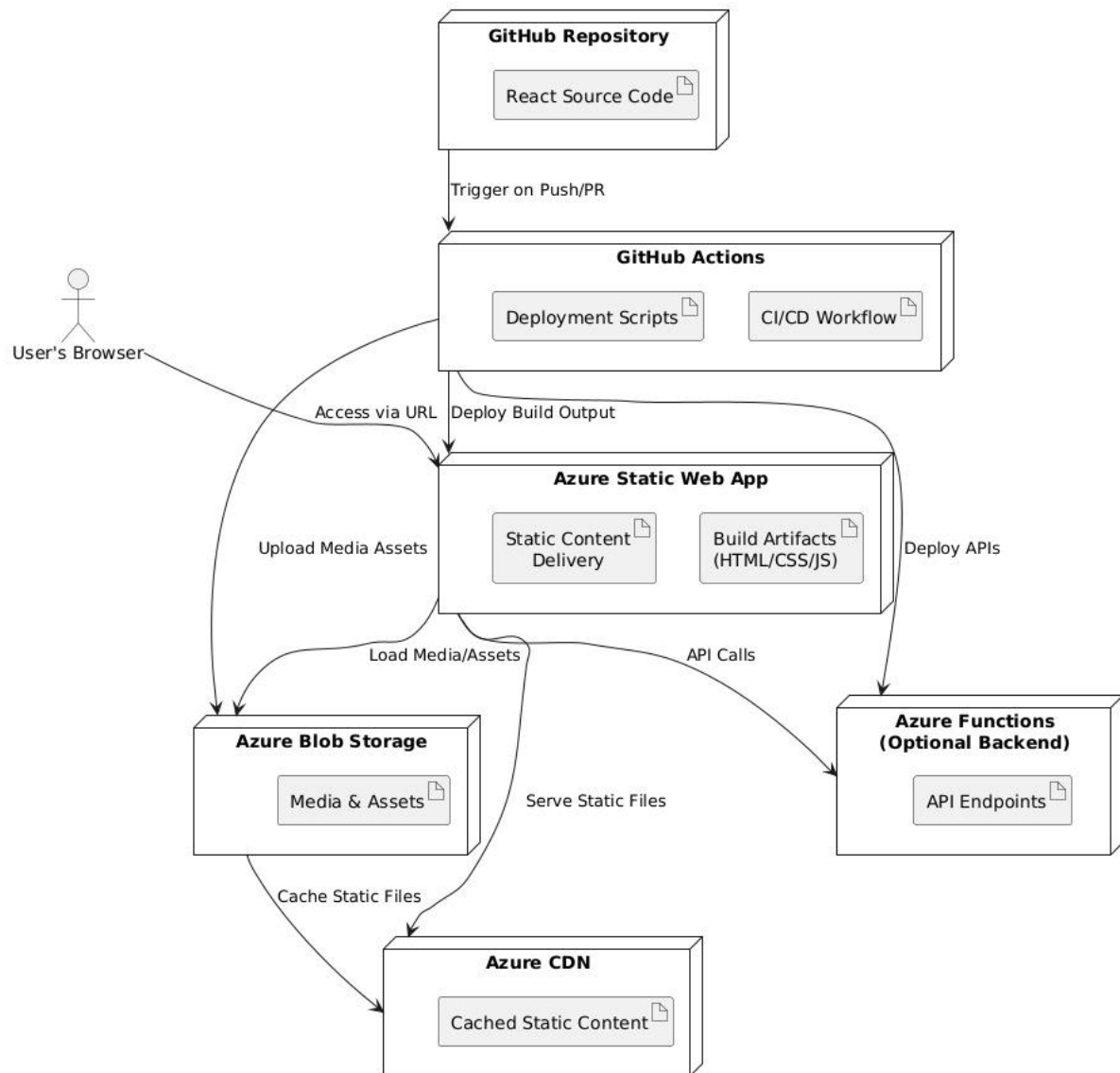
UML DIAGRAMS

**As a researcher, i should be able
to create a research project bifiln details**

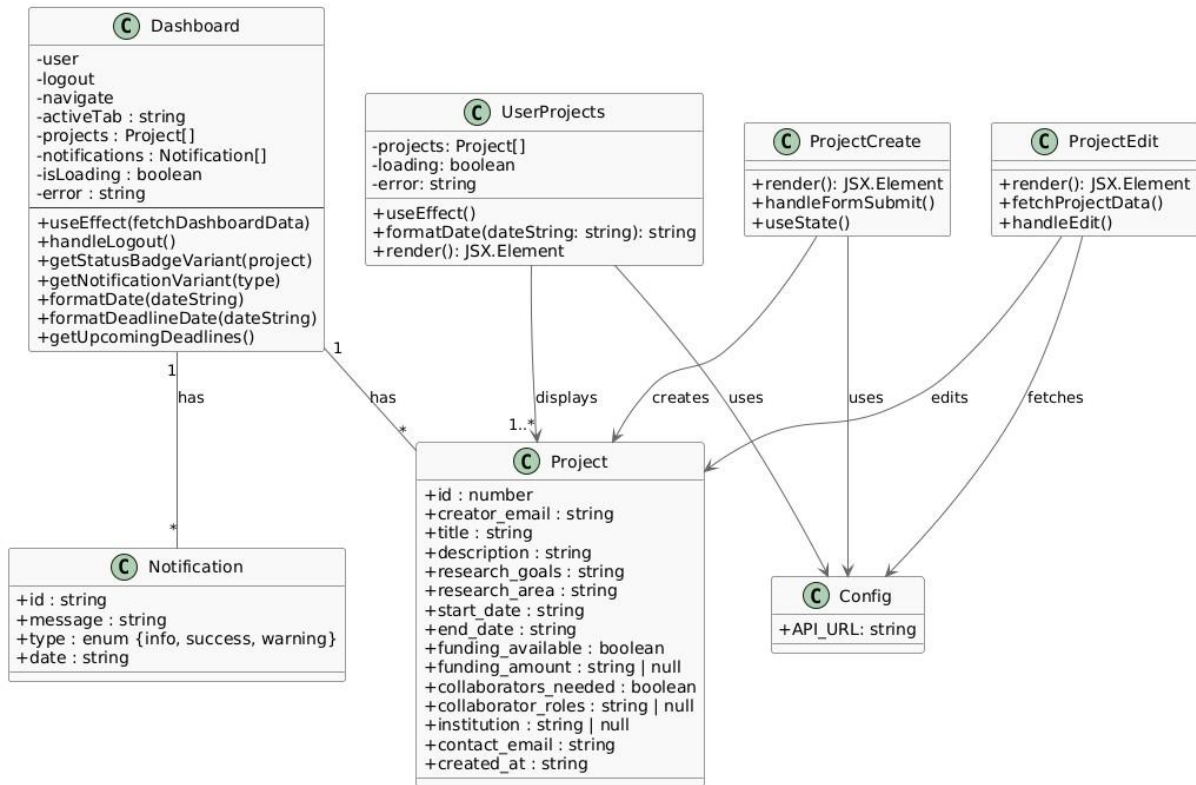


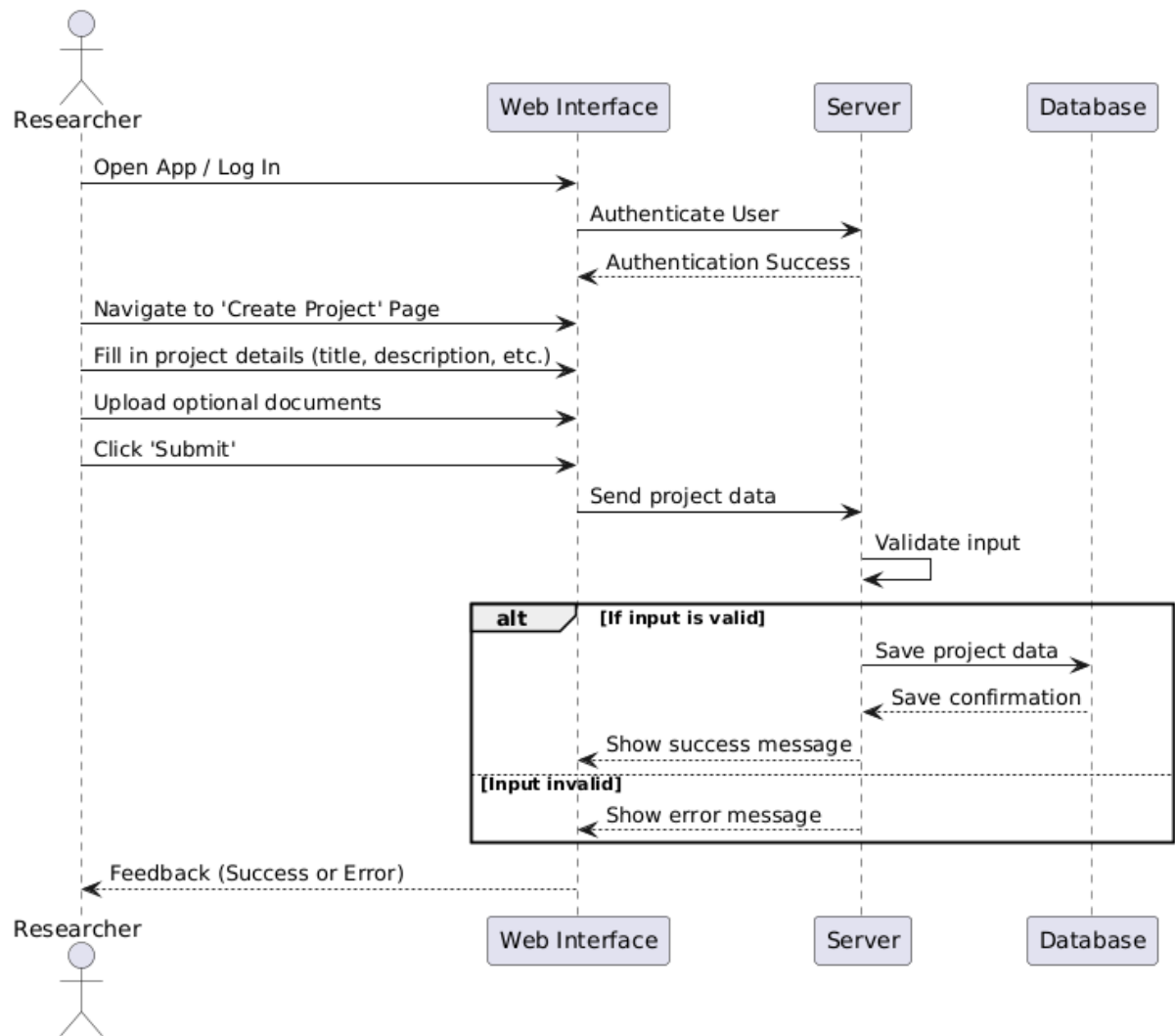
ResearchCollab Platform Use Case Diagram





ResearchBridge Dashboard - Class Diagram





Minute Meetings

Daily Scrum

Daily Scrum: 15 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange

- **What was done yesterday:**
 - Project scope discussed and responsibilities assigned.
 - GitHub repo created by Bongumusa.
 - Initial database schema brainstorming using MongoDB (planned).
 - Gabriel started Google authentication setup.
 - **Today's Plan:**
 - UI wireframes begin (Nyeleti, Hluma).
 - Nhlamulo to draft initial schema for MongoDB.
 - Gabriel to continue OAuth integration.
 - Documentation file structure set up by Nyeleti.
 - **Impediments:**
 - Team illness and religious observances affecting availability.
 - Internet interruptions for some members.
-

Daily Scrum: 16 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

- **What was done yesterday:**
 - MongoDB integration attempted.
 - OAuth login flow started.
 - Initial documentation sections drafted.
- **Today's Plan:**
 - Start converting UI wireframes into actual components.
 - Refine MongoDB schema logic (Nhlamulo).
 - OAuth redirect handling by Gabriel.
 - Begin test scaffolding (Tinashe, Bongumusa).
- **Impediments:**
 - MongoDB connection and schema modeling issues.

- Health and productivity still affected.
-

Daily Scrum: 17 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

- **What was done yesterday:**
 - Functional UI pages in progress.
 - Database issues with MongoDB led to reconsideration.
 - Google OAuth redirect bug encountered.
 - Test cases drafted with low success.
 - **Today's Plan:**
 - Explore alternatives to MongoDB due to performance issues.
 - Improve UI logic and form validation (Nyeleti, Hluma).
 - Tinashe to enhance coverage
 - Nhlamulo to research FreeSQLDatabase as replacement.
 - **Impediments:**
 - MongoDB integration unreliable.
 - Illness and religious commitments continue to impact pace.
 - Test coverage very low (<12%).
-

Daily Scrum: 21 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

- **What was done over the weekend:**
 - Light work due to religious holiday and health.
 - Decision made to switch from MongoDB to FreeSQLDatabase.
 - Initial connection tests with FreeSQLDatabase successful.
- **Today's Plan:**
 - Redesign schema for SQL (Nhlamulo).

- Adjust backend endpoints to match new database.
 - Update frontend to ensure compatibility.
 - Bongumusa to handle Git updates and push new DB config.
 - **Impediments:**
 - Rewriting data logic from NoSQL to SQL is time-consuming.
 - Low test coverage still persistent.
-

Daily Scrum: 22 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Database structure successfully converted to FreeSQLDatabase.
- Integration with backend tested.

Today's Plan:

- Complete migration to FreeSQLDatabase on production.
- Finish funding and collaboration fields.
- Gabriel to finalize Google auth.
- Test coverage update by Tinashe and Bongumusa.

Impediments:

- SQL syntax bugs causing data validation issues.
 - CSS conflicts affecting UI polish.
-

Daily Scrum: 23 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

- **What was done yesterday:**
 - Finalized FreeSQLDatabase deployment.
 - Auth flow working with UI.
 - UML diagrams and documentation nearing completion.

- **Today's Plan:**
 - Tinashe and Bongumusa to focus on boosting code coverage using Jest.
 - Nyeleti to complete README and documentation.
 - Nhlamulo to test SQL queries for robustness.
 - Final UI review and touch-ups (Nyeleti).
- **Impediments:**
 - Test coverage still not hitting targets (~35%).
 - Minor issues with SQL joins and field validations.

SCRUM RETROSPECTIVE

Sprint Retrospective Meeting Minutes

Date: 14/04/25

Time: 17:00

Venue: MSL 005, University of the Witwatersrand

Attendees:

- Renkie Nyeleti Mkhize
- Hluma Nziweni
- Tinashe Nganadange
- Bongumusa Makhubu
- Gabriel Maponya
- Nhlamulo Mabunda

1. Purpose of the Meeting

The purpose of this sprint retrospective was to reflect on the previous sprint, evaluate what went well, identify challenges, and determine actionable improvements for the next sprint.

2. Discussion Points

a) User Stories

- The team acknowledged that the user stories created for the sprint were too generalized and not specific enough to guide implementation clearly.
- This misunderstanding affected overall clarity and progress during the sprint.

b) Time Management and Task Execution

- The team faced challenges in balancing all aspects of the sprint within a week.
- A significant portion of the sprint was spent on debugging, leaving limited time for actual implementation.
- Assigning individual members to specialized roles led to inefficiencies, as some tasks were too large or complex for one person to handle alone.

c) GitHub and Deployment

- The GitHub repository and webpage deployment were successfully managed and are considered a positive outcome of the sprint.

d) Scrum Practices and Documentation

- There was poor documentation of the sprint and scrum meetings.
- Daily scrum meetings were often merged with other SCRUM ceremonies, which reduced their effectiveness and made it difficult to track progress clearly.

e) Team Communication

- It was noted that better communication was needed.
- Some team members struggled with issues silently and only raised them when it was too late for effective support or collaborative debugging.
- This delayed progress and added pressure late in the sprint.

3. Action Items and Areas for Improvement

- **Improve User Stories:**
Ensure user stories are specific, actionable, and well-understood by the entire team.
- **Better Task Allocation:**
Distribute tasks more evenly to avoid overburdening individuals and to encourage collaborative work.
- **Enhance Time Management:**
Strike a better balance between debugging and actual implementation to maintain productivity.

- **Improve Scrum Practices:**
Separate daily scrums from other ceremonies and ensure consistent documentation of all sprint-related meetings.
 - **Foster Open Communication:**
Encourage team members to speak up earlier when they encounter problems to enable quicker resolutions and shared learning.
-

Meeting Adjourned

SCRUM PLANNING MEETING

SCRUM Planning Meeting Minutes

Date: 14 April 2024

Time: 16:00

Venue: MSL 111, University of the Witwatersrand

Attendees:

- Renkie Nyeleti Mkhize
 - Hluma Nziweni
 - Tinashe Nganadange
 - Bongumusa Makhubu
 - Gabriel Maponya
 - Nhlamulo Mabunda
 - **Product Owner: Thisra Pillay**
-

1. Purpose of the Meeting

To plan and outline the scope of the upcoming sprint, based on functionality requirements shared by the Product Owner, including user stories and features for the researcher collaboration platform.

2. Product Owner Requirements

The Product Owner, **Thisra Pillay**, outlined **six key user stories** that should be covered during the sprint:

User Story 1: Create and Publish a Project Listing

A researcher should be able to create and publish project listings so that they can collaborate with other researchers.

Form Requirements:

- Project Title
 - Project Description
 - Research Goals (e.g., “I want to create a diffusion model”)
 - Key Research Areas (e.g., AI)
 - Start Date
 - End Date
 - Funding Available (Checkbox: Yes/No)
 - Researcher Details (Who’s looking for a collaborator)
 - Skills and Expertise
 - Experience Level: Beginner / Intermediate / Expert
 - Desired Positions: Project Manager, Lab Technician, Research Assistant
 - Technical Requirements
 - **Verification Step on Publish**
-

User Story 2: Edit a Project Listing

A researcher should be able to edit their existing project listings.

- Accessible via “My Projects”
- View all projects
- Click individual listing → View full details
- **Edit Button** next to each listing
- All listing details should be editable
- Changes can be saved

User Story 3: Delete a Project Listing

A researcher should be able to delete a project listing.

- “Delete” button next to “Edit”
- When clicked → Confirmation popup: “Are you sure?”

User Story 4: Invite Collaborators to a Project

A researcher should be able to invite collaborators to their project.

- Within each listing → “Invite Collaborators” button
- Search collaborators by: Name, Skill Set, or Position
- Show results → Select person → Invitation popup
 - Text field: “This is my project, I need xyz person to help.”
 - Send invitation

User Story 5: Manage Sent Invitations

A researcher should be able to manage all sent invitations.

- In each listing → “Manage Invitations” section
- Should display:
 - Who was invited
 - Role invited for
 - Status: Pending, Declined, Expired, Cancelled
 - Date invitation was sent

User Story 6: Respond to Received Invitations

A researcher should be able to view and respond to received invitations.

- Dashboard → “View Invitations” button
- Leads to a page showing all received invitations
- View full details

- Respond with: Accept / Decline
 - Invitation statuses: Accepted, Declined, Cancelled, Expired
-

3. Next Steps / Sprint Focus

- Finalize and refine the six user stories.
 - Begin by creating the UI for project creation.
 - Set up the database schema to support all listing fields and invitation tracking.
 - Assign tasks evenly among team members to prevent overload.
-

Meeting Adjourned

SPRINT REVIEW

Sprint Review Minutes

Date: 23/04/2025

Facilitator: Thisra Pillay (Product Owner)

Time: 16:00

Venue: MSL 111, University of the Witwatersrand

Attendees:

- Thisra Pillay (Product Owner)
 - Renkie Nyeleti Mkhize
 - Hluma Nziweni
 - Bongumusa Makhubu
 - Gabriel Maponya
 - Nhlamulo Mabunda
 - Tinashe Nganadange
-

Completed Work:

- **Login and Logout Functionality** successfully implemented and tested.
 - **Role Selection Page** is fully functional.
 - Users can now choose their roles as **Researcher**, **Reviewer**, or **Admin**.
-

Work In Progress / Not Completed:

- **Landing Page** was not completed.
 - This page was intended to provide basic details about the application.
 - **Google Authentication** is partially working.
 - Still requires refinement and full integration.
 - **Testing and Code Coverage** was not completed.
 - Code coverage tests were not run, and overall testing remains incomplete.
-

Issues & Observations:

- **Project Polish & UI Design:**
 - Overall UI is **not yet up to standard**.
 - Requires improvement to meet usability and visual quality expectations.
-

Feedback Summary:

- The team acknowledged steady progress on core functionality.
 - Stakeholders emphasized the importance of improved UI/UX moving forward.
 - Appreciation for completed features was expressed, with encouragement to maintain momentum.
-

Next Steps:

- Finalize the **Landing Page**.
- Complete full integration of **Google Authentication**.
- Improve overall **UI polish and user experience**.

- Develop additional **core functionalities** based on the product roadmap (e.g., dashboard features, project creation flow, collaboration tools, etc.).
 - Prioritize **testing and code coverage** to ensure product stability and reliability.
-

Meeting Adjourned

Backlog Refinement

Backlog Refinement Meeting: 16 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange

- **Review of Backlog Items:**
 - Reviewed all existing user stories and tasks. Some of the tasks related to the database were found to be outdated (MongoDB-related), so those were marked for updates or removal.
 - Decided to add a task to switch the database from MongoDB to FreeSQLDatabase, and further refine the schema.
- **Prioritization:**
 - High priority was given to tasks related to UI refinement (since it affects user interaction) and the database switch.
 - Medium priority tasks included finalizing Google OAuth integration and preparing test cases.
- **Clarification of User Stories:**
 - The story regarding user authentication was split into smaller stories for handling different OAuth flow cases (login, signup).
 - The database story was split into sub-tasks: Database migration, schema redesign, and API changes.
- **Re-estimating Work:**
 - Re-estimated tasks related to Google OAuth integration (was originally estimated at 5 hours, now 8 hours due to issues with redirects).

- Database migration tasks were estimated to take 3 additional days due to the switch from MongoDB to FreeSQLDatabase.
 - **Preparation for Sprint:**
 - The top priority tasks for the next sprint were confirmed: UI components, database migration, and authentication.
 - **New Items Added to Backlog:**
 - Added task for deploying FreeSQLDatabase and configuring the connection for the frontend.
 - Added task for creating detailed test cases for the new database structure.
-

Backlog Refinement Meeting: 21 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange

- **Review of Backlog Items:**
 - All items related to MongoDB were updated or removed, and FreeSQLDatabase tasks were added.
 - New UI components related to research project creation were reviewed and clarified.
 - Backlog items for Google OAuth were marked as complete, and it was agreed that they were ready for testing.
- **Prioritization:**
 - High priority tasks included completing the database migration, setting up proper test coverage, and finalizing UI components.
 - Medium priority was given to documentation and research project categorization.
- **Clarification of User Stories:**
 - User stories for database schema adjustments were clarified, and acceptance criteria were added.
 - Test case coverage for database queries was clarified to include edge cases.

- **Re-estimating Work:**

- Database migration tasks were re-estimated to take 2 more days due to unforeseen complexity in the data structure change.
- Frontend tasks for UI refinements were re-estimated at 3 more days due to CSS conflicts.

- **Preparation for Sprint:**

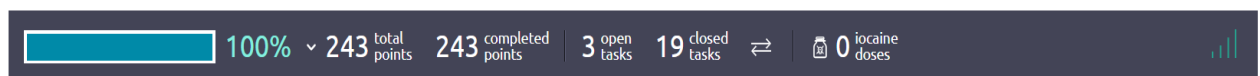
- Tasks that were ready for the next sprint were confirmed: Database migration, UI refinements, and integration of Google OAuth.
- Some backlog items were deferred to a future sprint due to dependencies on the above tasks.

- **New Items Added to Backlog:**

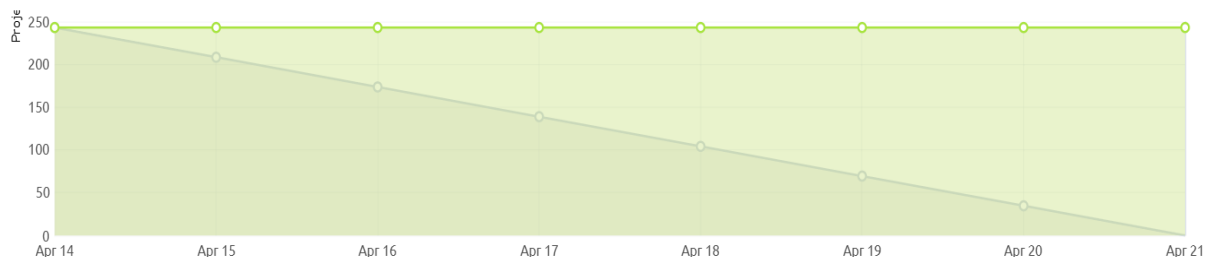
- Added task for improving code coverage with Jest and ensuring test cases are comprehensive.
- Added task for fixing deployment-related bugs encountered with FreeSQLDatabase.

Sprint Burndown Chart

Sprint 2 14 Apr 2025 to 21 Apr 2025



How this chart works



Sprint 3

Sprint 3: Feature Implementation Summary

Deliverables

- Admin & Reviewer Workflows
 - Proposal Evaluation Flow
 - Internal Messaging System
-

Updated User Stories with Functional Requirements

1. Admin User Management

As an admin,

I need to oversee researcher and reviewer accounts and assign roles as needed,

So that the platform remains secure, organized, and properly managed.

Functional Requirements:

- a. The admin should have access to a full list of all registered platform users, displaying each user's current role (e.g., Researcher, Reviewer).
 - b. The admin can edit user roles—e.g., upgrade a Researcher to a Reviewer when necessary.
-

2. Admin View of Submitted Proposals

As an admin,

I want to access all research submissions from a central dashboard,

So that I can efficiently manage the review process and assign appropriate reviewers.

Functional Requirements:

- a. Upon logging in, the admin is directed to the "**Submitted Proposals**" dashboard.
 - b. Admins can click on any individual proposal to see:
 - Full proposal details
 - A summary of its key research themes (populated from project creation info)
-

3. Assigning Reviewers to Proposals

As an admin,

I need to allocate reviewers to each research submission,

So that proposals are assessed by qualified individuals in an organized and timely way.

Functional Requirements:

- a. Each proposal page includes an **"Assign Reviewer(s)"** button.
 - b. Clicking the button opens a selection interface where admins can:
 - Browse and filter potential reviewers by expertise (from user profiles)
 - Assign one or more reviewers to the proposal
-

4. Reviewer Dashboard

As a reviewer,

I want a personalized dashboard listing my assigned proposals,

So that I can easily manage and access the submissions I need to evaluate.

Functional Requirements:

- a. Upon logging in, the reviewer is shown a dashboard titled **"Assigned Proposals"**.
 - b. Clicking a proposal provides full access to its details and attached documents.
-

5. Reviewing and Evaluating Proposals

As a reviewer,

I want to provide structured feedback on each assigned proposal,

So that I can help determine whether it should be accepted, revised, or rejected.

Functional Requirements:

- a. Each proposal page has an **"Evaluate"** button that opens a form.
 - b. Reviewers can:
 - Input written feedback
 - Choose an outcome (Approve, Revise, Reject)
 - Submit their evaluation
 - c. Once submitted, the evaluation result should reflect in the proposal's review status.
-

6. Internal Messaging System

As a user (admin, researcher, or reviewer),

I want to communicate directly within the platform,

So that I don't need to rely on external tools for collaboration and coordination.

Functional Requirements:

a. All users can access a **messaging inbox** from their dashboard.

b. Users can:

- Search and select message recipients (by name or role)
- Write a message with a subject and body

Optionally attach files or link to proposals/projects

c. In-app notifications alert users of new messages.

d. Users can reply to ongoing conversations

User Stories and User Acceptance tests

1. Admin Role Management

User Story (Who-What-Why):

As an Admin, I want to manage researcher and reviewer accounts by assigning and editing roles, so that the platform remains organized and secure.

User Acceptance Tests:

- **Given** I am logged in as an Admin,
When I navigate to the user management page,
Then I should see a list of users and their current roles.
- **Given** I am on the user management page,
When I select a user and choose "Edit Role",
Then I should be able to assign them a new role (e.g., Reviewer).

2. Admin Proposal Dashboard

User Story (Who–What–Why):

As an Admin, I want to view all submitted research proposals in a dashboard, so that I can manage and assign them effectively.

User Acceptance Tests:

- **Given** I am logged in as an Admin,
When I open the dashboard,
Then I should see a list of all submitted proposals.
 - **Given** I have a list of proposals,
When I click on one,
Then I should be able to view its full details including research concepts.
-

3. Reviewer Assignment**User Story (Who–What–Why):**

As an Admin, I want to assign reviewers to research proposals, so that proposals are reviewed by suitable individuals.

User Acceptance Tests:

- **Given** I am viewing a proposal,
When I click “Assign Reviewer”,
Then I should be able to select a reviewer based on expertise.
 - **Given** I’ve selected a reviewer,
When I confirm the assignment,
Then the reviewer should be linked to that proposal in the system.
-

4. Reviewer Dashboard**User Story (Who–What–Why):**

As a Reviewer, I want to view a dashboard listing proposals assigned to me, so that I can manage and access them easily.

User Acceptance Tests:

- **Given** I am logged in as a Reviewer,
When I open my dashboard,
Then I should see a list of proposals assigned to me.
- **Given** I have assigned proposals,
When I click on a proposal from the list,
Then I should be taken to the detailed view of the proposal.

5. Reviewer Evaluation

User Story (Who–What–Why):

As a Reviewer, I want to evaluate assigned proposals and submit feedback, so that I can contribute to the review process.

User Acceptance Tests:

- **Given** I am viewing a proposal,
When I click on “Evaluate”,
Then I should be able to enter feedback and select an evaluation outcome (approve, revise, reject).
 - **Given** I submit my evaluation,
When I confirm it,
Then the feedback should be stored and visible in the proposal details.
-

6. Messaging System

User Story (Who–What–Why):

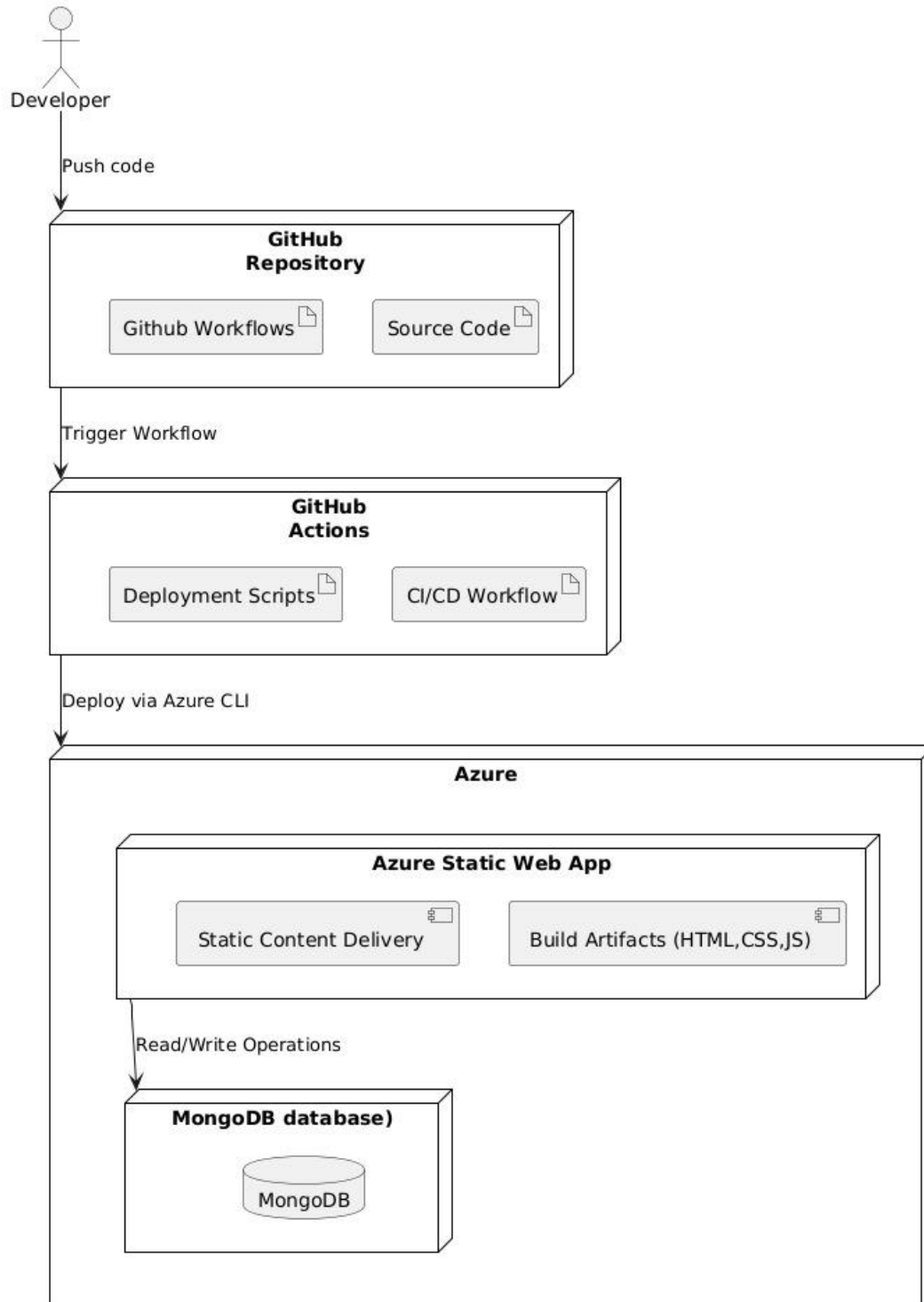
As a user (Admin, Reviewer, or Researcher), I want to send and receive messages, so that I can communicate with others on the platform.

User Acceptance Tests:

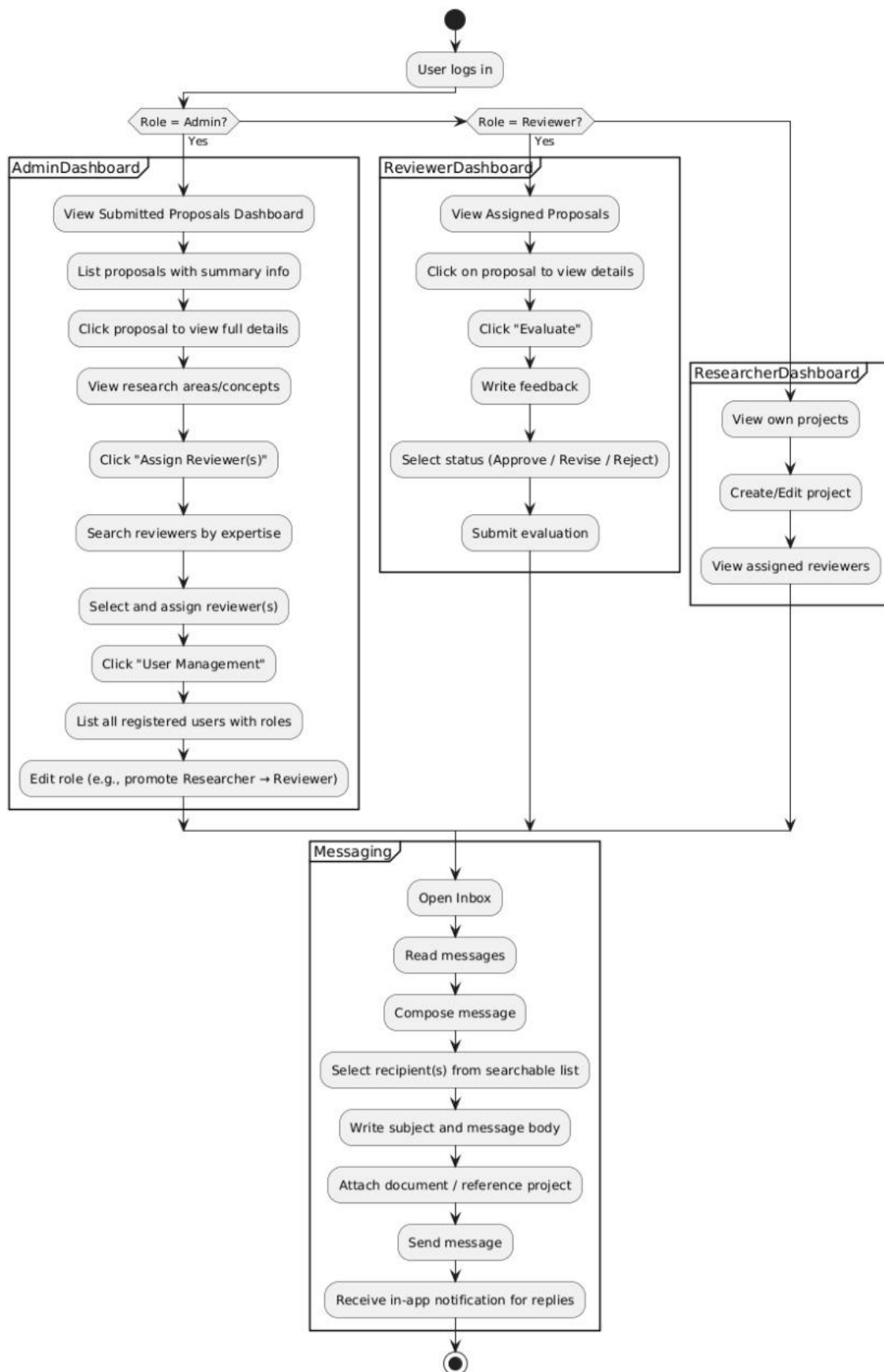
- **Given** I am logged in,
When I open the messaging feature,
Then I should see my inbox with received messages.
 - **Given** I want to send a message,
When I search for a user and write a message,
Then the message should be delivered and appear in their inbox.
 - **Given** I receive a message,
When I read it,
Then a notification should disappear and it should be marked as read.
-

UML DIAGRAMS

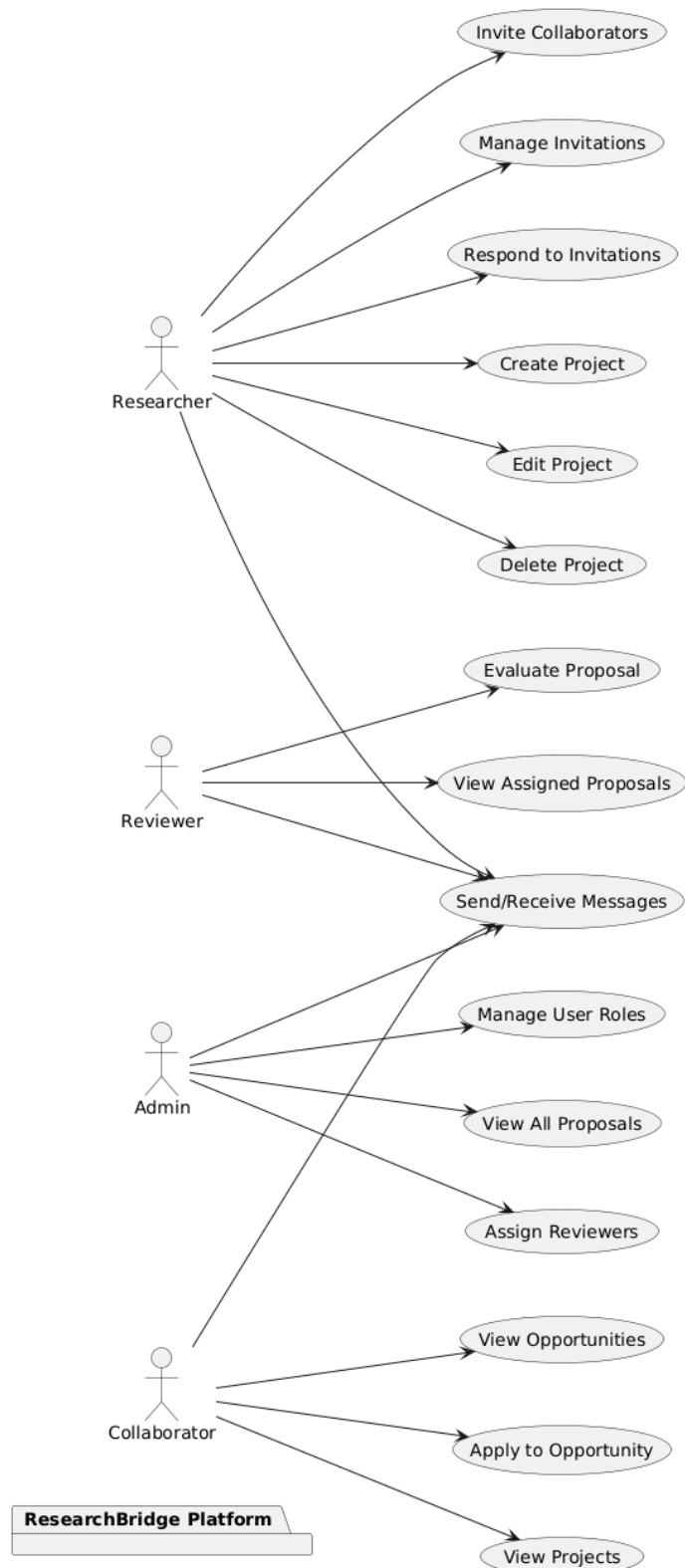
Deployment Diagram



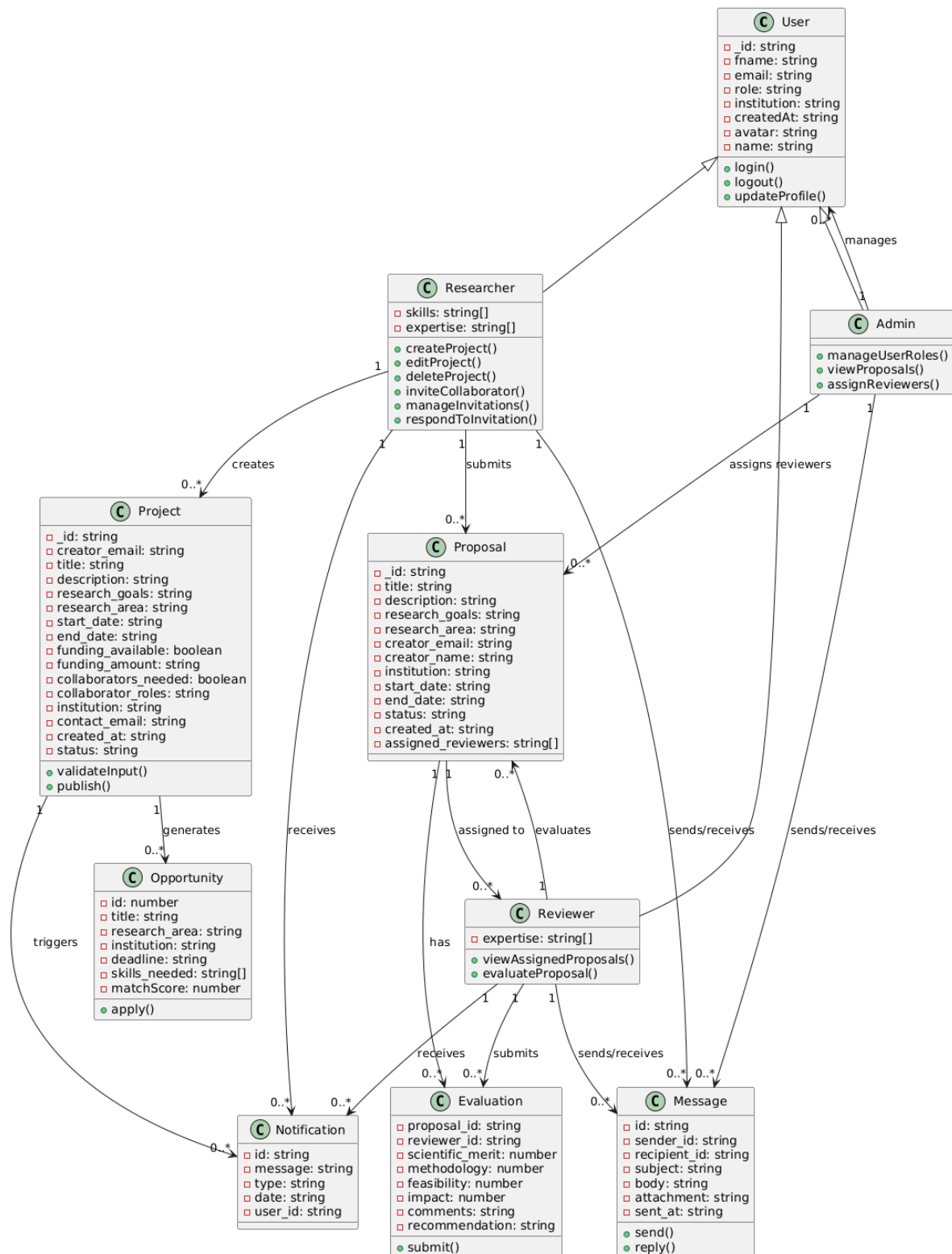
Activity diagram



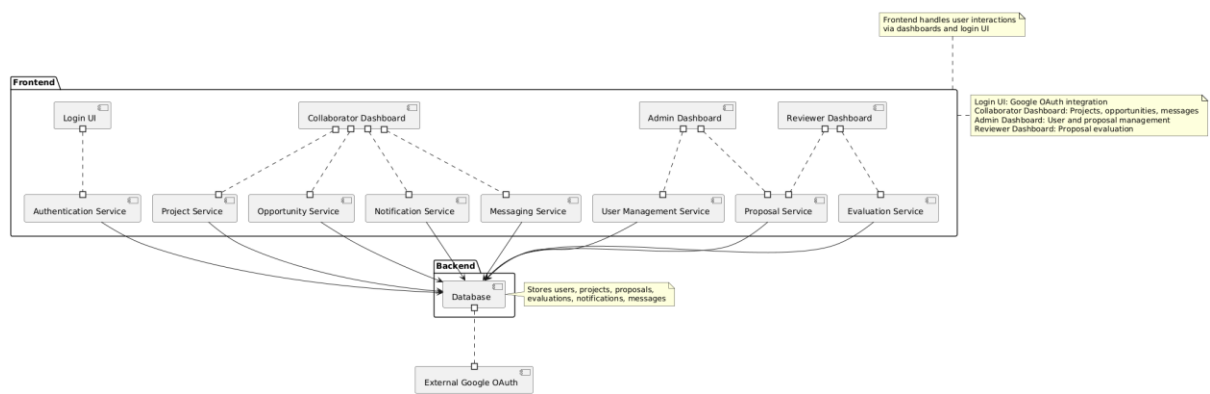
Use Case Diagram



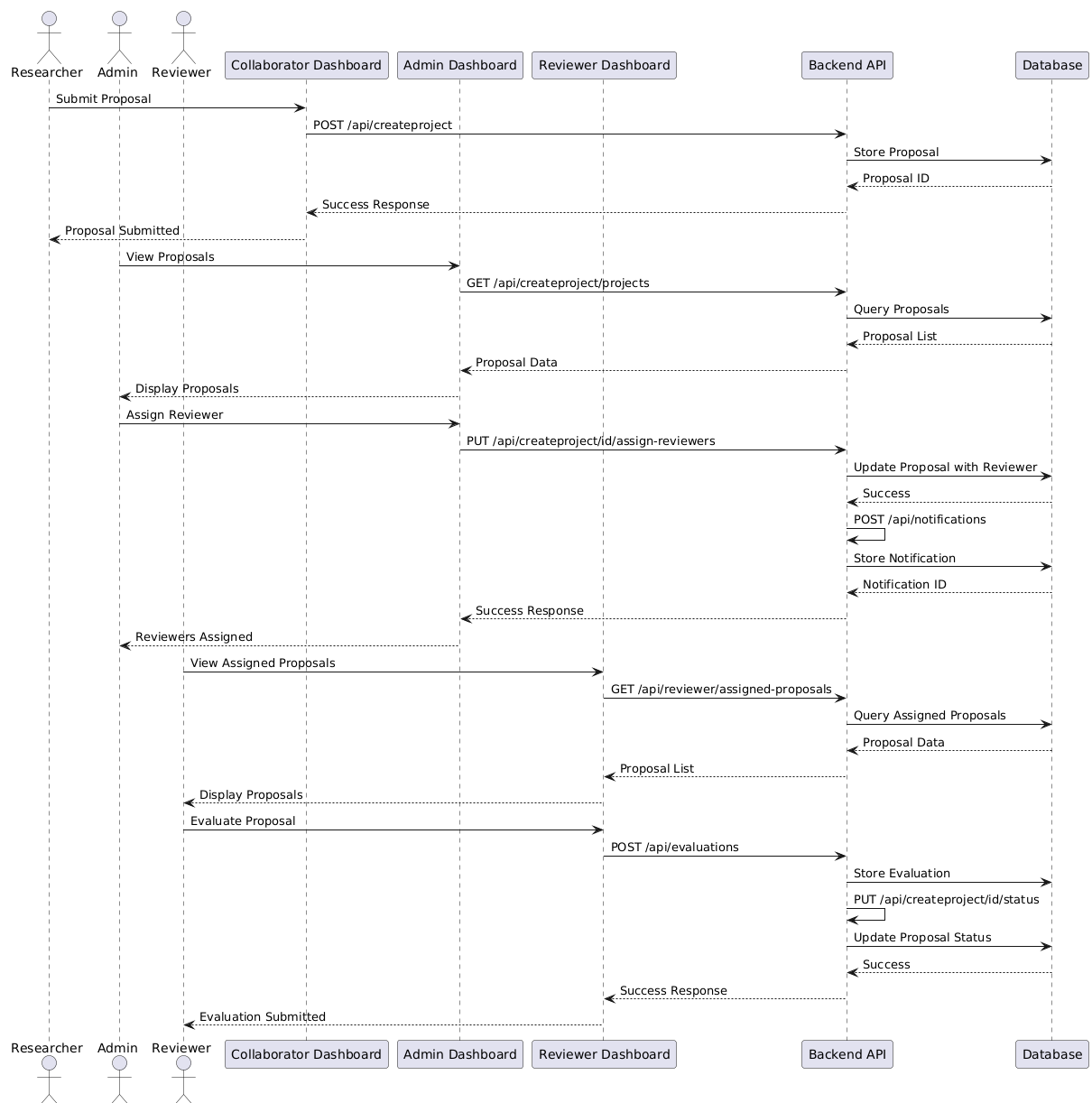
Class Diagram



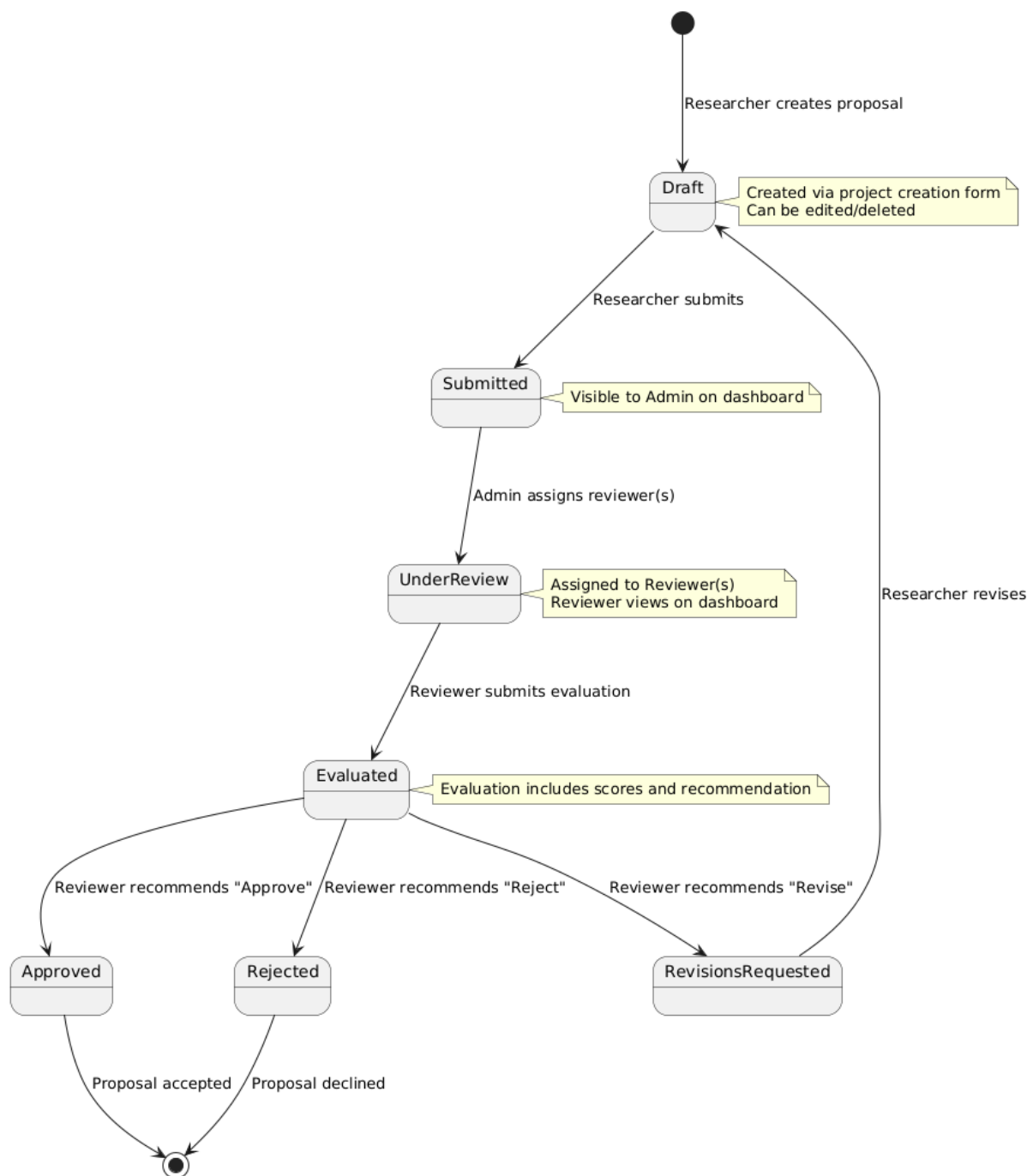
Component Diagram



Sequence Diagram



State Diagram



DAILY SCRUM

Daily Scrum: 25 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Tinashe Nganadange, Nhlamulo Mabunda

What was done yesterday:

- Sprint Review with Refiloe Mopeloa (client).
- Clarified MVP: 5 meetings, UML documentation, 6 user stories, GitHub release, 60% test coverage.
- Gabriel started Google auth setup.
- Hluma & Tinashe worked on nav bar.
- Nhlamulo began MongoDB migration.
- Nyeleti worked on UML Use Case View.
- Musa set up test framework.

Today's Plan:

- Link navigation views.
- Gabriel to handle role-based routing.
- Nyeleti to map UML views to user stories.
- Musa to test login & messaging.
- Hluma to polish UI.
- Nhlamulo to define MongoDB user roles.

Impediments:

- Role-based dashboard routing unclear.
- MongoDB schema incomplete.
- Adjusting to sprint expectations.

Daily Scrum: 28 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Calendar component started (Tinashe).
- Session persistence added (Gabriel).
- Login & SignUp tests done (Musa).
- Messages UI built (Hluma).

- UML views created for Admin Role Mgmt (Nyeleti).
- MongoDB users collection setup (Nhlamulo).

Today's Plan:

- Gabriel to fix dashboard redirection.
- Musa to write Calendar & Notification tests.
- Nyeleti to build Reviewer dashboard UML & sequence diagrams.
- Tinashe to add delete to Notifications.
- Nhlamulo to support notification schema.

Impediments:

- Notification/message routing bugs.
- UI changes breaking tests (Musa).
- Proposal schema issues in DB (Nhlamulo).

Daily Scrum: 29 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Fixed login redirection.
- Calendar & Chat tested (Musa).
- CollaboratorDashboard started (Tinashe).
- UML aligned with user stories 4–6 (Nyeleti).
- Proposal data migrated to MongoDB (Nhlamulo).

Today's Plan:

- Finish dashboard routing.
- Implement reviewer assignment logic.
- Trigger GitHub release on push to main.
- Reach 40% coverage (Musa).
- Continue updating 4+1 views (Nyeleti).

Impediments:

- Reviewer dashboard not loading content.
 - Flaky CodeCov reports.
 - DB read issues.
-

Daily Scrum: 30 April 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Dashboard routing done.
- Reviewer Evaluation UI built.
- Messaging & Notification tests passed (Musa).
- Logical UML view completed.
- DB holds researcher-reviewer links.

Today's Plan:

- Full CRUD on proposals/notifications.
- Finalize session tokens.
- Target 55% test coverage.
- UML Development View completion.
- Dashboard content to query DB.

Impediments:

- Role-filtered data inaccurate.
 - Notification delete error.
 - UML assumptions breaking due to UI.
-

Daily Scrum: 1 May 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Login auth secured (Gabriel).
- Reached 58% code coverage.
- Proposal data loaded on all dashboards.
- UML Process & Deployment views done.
- Notification delete fixed.

Today's Plan:

- Push main + verify GitHub release.
- Refine tests for Messaging & Calendar.
- DB optimization for Reviewer filters.
- UML PDF compilation.

Impediments:

- Message replies vanish after reload.
- Tests fail in production build.
- Diagram mismatches.

Daily Scrum: 2 May 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: All present

What was done yesterday:

- Dashboard routing finalized.
- All user stories connected.
- UML completed & submitted.
- GitHub release created.
- 60% coverage reached.

Today's Plan:

- Final cleanup and bug fixes.
- Sprint summary and demo prep.
- Finalize 4+1 to match MVP.

Impediments:

- Notification responsiveness issues.
 - In-app messaging not syncing deletes.
-

SPRINT REVIEW

Sprint Review Meeting

Date: 24 April 2025

Time: 16:30

Location: 104 MSL LAB, University of the Witwatersrand

Facilitator: Renkie Nyeleti Mkhize

Client: Refiloe Mopeloa

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Tinashe Nganadange, Nhlamulo Mabunda

Completed Work:

- **Landing page** was designed and implemented with a clean, responsive layout.
 - **Researchers** can now **create project listings**.
 - **Edit functionality** for project listings has been successfully integrated.
 - General **polish and UI/UX refinements** were made for better user interaction.
-

Work in Progress:

- **Login, SignUp**, and **Google Authentication** components are nearing completion.
 - Researcher functionality to **delete project listings** is under development.
 - Feature to allow **researchers to find collaborators** is being built.
 - Displaying **deadlines for projects** on dashboards is in progress.
 - Ongoing **transition of the database** to improve performance and structure.
-

Issues and Observations:

- **Deployment was problematic**, causing delays and missed test windows.
 - **Integrated testing remains unfinished**, lowering confidence in feature stability.
 - **Initial user stories lacked clarity**, requiring refinement mid-sprint.
 - **Frontend-backend coordination** was slowed by differing data structure expectations.
 - Time was lost adapting to **changes in the database schema** during transition.
-

Client Feedback:

- Refiloe was pleased with the visual quality of the **landing page** and ability to create/edit listings.
 - She emphasized the need for **accurate deadline visibility** and smoother **collaborator search**.
 - She highlighted concerns about **deployment reliability** and **test coverage**.
-

Action Items for Next Sprint:

- Finalize and deploy **authentication (login, signup, Google)**.
 - Complete **delete listing** and **collaborator search** functionality.
 - Finish and stabilize **integrated testing** using Jest.
 - Revise and finalize **user stories** with more accurate acceptance criteria.
 - Ensure full **database migration** is completed and aligned with frontend.
 - Improve **deployment pipeline** and fix reliability concerns.
-

BACKLOG REFINEMENT

Backlog Refinement Meeting: 5 May 2025

Facilitator: Renkie Nyeleti Mkhize

Attendees: Renkie Nyeleti Mkhize, Gabriel Maponya, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange

Review of Backlog Items:

- Reviewed remaining backlog items from the previous sprint.
 - Confirmed completion of landing page and researcher listing/edit functionality.
 - Noted that login, signup, and Google authentication tasks are in progress.
 - Found several items (e.g., collaborator feature, delete listing) lacked clear acceptance criteria and were marked for refinement.
 - Tasks related to FreeSQLDatabase were reviewed; flagged for transition to MongoDB.
-

Prioritization:

- High priority: Completing authentication flow (login/signup/Google), delete listing feature, and database transition.
 - Medium priority: Collaborator search feature, project deadline display, and frontend UI refinements.
-

Clarification of User Stories:

- Refined the “delete listing” story to include success/failure feedback and frontend confirmation modals.
 - Clarified the “find collaborators” story by defining filters and search functionality.
 - The deadline display story was adjusted to include both frontend logic and backend validation.
 - Database transition was split into schema redesign, data migration, and endpoint adjustments.
-

Re-estimating Work:

- Authentication stories extended by 2 more days due to redirect issues and token handling.
- Project deletion feature re-estimated due to backend route limitations (additional 6 hours).
- Collaborator feature re-estimated based on required search/filtering logic (2 more days).

Preparation for Sprint:

- Confirmed the sprint will focus on stabilizing user authentication, database migration, and delete functionality.
- Tasks were updated in Taiga with new subtasks and clearer labels (frontend/backend/testing).
- Testing and deployment tasks were marked for review at end of sprint, to catch up on unfinished integration testing.

New Items Added to Backlog:

- Added: Task for integrating collaborator search UI with backend filters.
- Added: Task for creating UI indicators for project deadlines (e.g., color coding, countdown badges).
- Added: Task to revise outdated acceptance criteria and link all user stories to testing tasks.

SCRUM PLANNING

Scrum Planning Meeting

Date: 5 May 2025

Time: 16:30

Venue: University of Witwatersrand, Lab 104

Facilitator: Product Owner – Refiloe Mopeloa

Attendees:

- Hluma Nziweni
- Gabriel Maponya
- Bongumusa Makhubu
- Tinashe Nganadange
- Renkie Nyeleti Mkhize
- Nhlamulo Mabunda

Purpose of the Meeting

To define and plan the scope of Sprint 3, ensuring that the team fully understands the user stories, technical priorities, and required actions to deliver according to the product owner's vision.

Sprint 3 Focus: Product Owner Expectations

Refiloe Mopeloa outlined the following **user stories** for the sprint:

1. Admin Role Management

- **User Story:** *As an Admin, I want to manage researcher and reviewer accounts, including assigning roles, so that the platform is organized and secure.*
 - **Key Requirements:**
 - Ability to view all users and their respective roles.
 - Functionality to edit roles (e.g., promote a researcher to a reviewer).
-

2. Admin Proposal Dashboard

- **User Story:** *As an Admin, I want to view all submitted research proposals in a dashboard, so that I can assign reviewers and manage the review process.*
 - **Key Requirements:**
 - Dashboard opens to "Submitted Proposals".
 - Option to click and view detailed proposal information and research concepts.
-

3. Reviewer Assignment

- **User Story:** *As an Admin, I want to assign reviewers to proposals, so that proposals are evaluated by qualified people in a timely, organized way.*
- **Key Requirements:**
 - Display proposals individually.
 - Incorporate an "assign reviewer" button with the ability to select reviewers based on expertise.

4. Reviewer Dashboard

- **User Story:** *As a Reviewer, I want to see a dashboard listing proposals assigned to me, so that I can easily manage and access them.*
- **Key Requirements:**
 - The dashboard should display “Assigned Proposals”.
 - Each proposal should be clickable to view its details.

5. Reviewer Evaluation

- **User Story:** *As a Reviewer, I want to evaluate assigned proposals, so that I can give structured feedback and recommendations (approve, revise, reject).*
- **Key Requirements:**
 - An “Evaluate” button that facilitates feedback submission.
 - Evaluation results should be reflected in the corresponding proposal's data.

6. Messaging System (for All Users)

- **User Story:** *As a user (Researcher, Reviewer, or Admin), I want to send and receive messages, so that I can communicate within the platform.*
- **Key Requirements:**
 - A messaging inbox for all users.
 - Ability to send messages (with features like a searchable list, subject/body fields, attachment support).
 - In-app notifications for new messages and capability to reply.

Action Items & Areas of Focus

- **Backlog Finalization:**
 - Clearly refine and update the Sprint 3 user stories in Taiga based on the detailed requirements.
- **Module Development:**

- Focus on building the Reviewer Dashboard and Admin Role Management functionalities.
 - Ensure the Admin Proposal Dashboard and Reviewer Evaluation features meet the defined needs.
 - **Testing:**
 - Establish integrated GitHub testing for new and existing features.
 - Set a milestone for reaching the targeted test coverage.
 - **Documentation and Diagrams:**
 - Update the UML diagrams (4+1 views) to reflect the new functional requirements.
 - Compile comprehensive documentation for each user story.
 - **Communication and Tracking:**
 - Maintain clear daily updates on Taiga.
 - Log any blockers immediately and address issues during daily scrums.
-

SCRUM RETROSPECTIVE

Scrum Retrospective Report

Date: 5 May 2025

Attendees: Hluma Nziweni, Gabriel Maponya, Bongumusa Makhubu, Tinashe Nganadange, Renkie Nyeleti Mkhize, Nhlamulo Mabunda

Facilitator: Team-led

Purpose of the Meeting:

To evaluate what went well, identify key challenges faced during the sprint, and determine actionable improvements for the next sprint.

1. Discussion Points

a) User Stories

- The team acknowledged that the initial user stories were too broad and general.
- As we progressed, the sprints became overpopulated with overlapping or unclear tasks.

- Feedback from the client (Refiloe Mopeloa) clarified that tasks confined to a single page or feature are not valid user stories.
- We need to focus on more outcome-driven stories that provide value to the user.

b) Role Reassignment

- The original model of assigning one role per member proved inefficient.
- Having a single person specialize in one task caused bottlenecks and delays.
- The team decided to adopt a more flexible, collaborative approach: individuals still lead their focus areas but actively contribute across multiple tasks.

c) Testing

- We recognized the need for integrated testing using GitHub.
- Issues were not being properly logged, tracked, or resolved on GitHub.
- Future sprints will emphasize continuous integration, GitHub Actions, and thorough documentation of errors.

d) UML Diagrams

- Most UML diagrams were well-received, except the Use Case diagram which was overcrowded.
- The visual size and clarity of diagrams need to be improved for better interpretation and client readability.

e) Project Polish

- Critical technical improvements are still pending: API routes, deployment process, and cleanup of misaligned or outdated user stories.
- These will be prioritized in the next sprint to ensure a smooth MVP handoff.

f) Communication

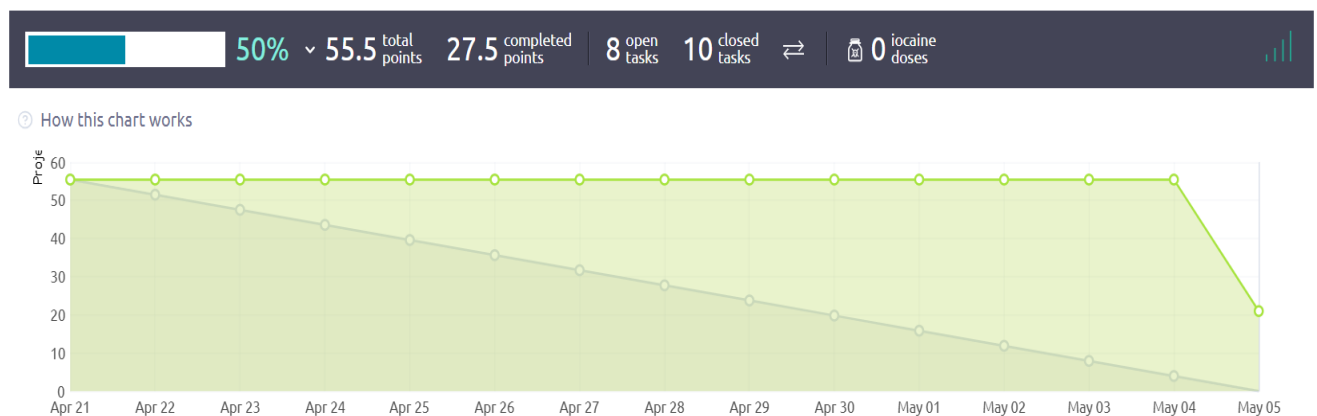
- Communication remains a key challenge.
- The team failed to consistently log tasks, blockers, and progress on Taiga.
- Updates were scattered or shared too late, limiting project visibility and coordination.

2. Action Items & Areas of Improvement

Area	Action Item	Responsible/All
User Stories	Refactor stories to be more specific, goal-oriented, and outcome-based.	All
Role Collaboration	Encourage pair programming, knowledge sharing, and cross-function support.	All
Testing	Implement integrated GitHub testing with logs; automate where possible.	Musa + All
UML Diagrams	Simplify Use Case diagrams; increase canvas size for readability.	Nyeleti
Project Polish	Clean API structure, fix broken deployments, finalize user story alignment.	Gabriel, Nhlamulo, Tinashe
Communication	Actively use Taiga for all tasks/issues and share daily updates.	All

Sprint Burndown Chart

Sprint 3 21 Apr 2025 to 05 May 2025



Sprint 4

1. Grant and Funding Tracking

As a researcher,

I want to monitor how my grants and funding are being spent,

So that I can manage my budget wisely and plan for future financial needs.

Functional Requirements:

a. A **“Funding”** tab should be available on the researcher's dashboard to access a dedicated funding management section.

b. In the funding dashboard, the researcher can initiate a new funding tracker and assign it to a specific project.

c. The tracker should allow input for:

- Grant Title
- Project Name
- Total Funding Awarded
- Amount Spent
- Remaining Balance
- Start Date
- Funding Status (e.g., Active, Inactive)

d. Clicking on a specific project opens a detailed funding page where the researcher can define **funding categories** such as:

- **Personnel** (e.g., assistants, collaborators)
- **Equipment** (e.g., sensors, machines)
- **Consumables**
- **Miscellaneous**

e. All financial data entries must be editable.

f. The main funding dashboard should display a visual chart (e.g., pie chart) summarizing the funding status of each project.

2. Milestone Management and Progress Tracking

As a researcher,

I want to create and update milestones for my projects,

So that I can monitor progress and ensure deadlines and deliverables are met.

Functional Requirements:

a. A **“Milestones”** tab should be added to the dashboard to navigate to the milestone tracking section.

b. On the milestone dashboard, the researcher can add new project timelines and assign them to specific research projects.

c. For each milestone, the researcher can specify:

- Title
- Description
- Expected Completion Date
- Status (Not Started, In Progress, Completed)
- Assigned Team Member

d. Researchers should be able to update, delete, or change the status of each milestone.

e. The dashboard must present a project-wise breakdown of milestone statuses with a pie chart indicating progress (e.g., % completed, in progress, not started).

3. PDF Report Exporting

As a researcher,

I want the ability to export my dashboards as PDFs,

So that I can easily share project insights, progress summaries, and funding reports with team members, stakeholders, and sponsors.

Functional Requirements:

a. All three dashboards—**Milestones**, **Funding**, and **Custom View**—should have an **Export as PDF** button.

b. Researchers should be able to generate reports for individual projects directly from each respective dashboard.

4. Customizable Researcher Dashboard

As a researcher,

I want to personalize my dashboard by choosing what information and visual elements appear,

So that I can focus on the metrics, tasks, and updates that matter most without distraction.

Functional Requirements:

- a. Upon logging in, researchers can click a **"My Dashboard"** button to access a customizable workspace.
 - b. In this section, researchers can click **"Add Widget"** and choose what they'd like to display via a dropdown menu with options like:
 - My Projects
 - Milestones
 - Funding
 - c. Each widget maintains its full functionality:
 - **My Projects:** Add/edit/delete listings, invite/manage collaborators
 - **Milestones:** Show summary tables; clicking a project leads to detailed tracking
 - **Funding:** Present funding overview tables; clicking opens the detailed budget breakdown
-

User Stories and User Acceptance Tests

A)

User Story:

As a researcher I would like to track my funding status, expenditures and grants in so doing manage the budget for the research wisely.

User Acceptance Test:

Given that I am a logged-in researcher with access to my project dashboard,

When I navigate to the funding section,

Then I should see detailed and up-to-date information about funding status, expenditures, and grants, enabling me to manage the research budget effectively.

B)

User Story:

As a researcher, I want to create and update project milestones, so that I can monitor the progress of my research and ensure that deliverables are completed on time.

User Acceptance Test:

Given that I am on the project milestones page of my research project,

When I add or update a milestone with a description and due date,

Then the milestone should be saved and reflected on the project timeline and progress indicators.

C)

User Story:

As a researcher, I want to customize which widgets, data sections, and visual components display on my dashboard, so I can focus on the metrics, tasks, and updates most relevant to my work while minimizing congestion.

User Acceptance Test:

Given that I am viewing my dashboard,

When I select and configure the widgets, data sections, or visual components I want to display,

Then the dashboard should update to reflect my preferences, hiding unselected elements and showing only the chosen components.

D)

User Story:

As a researcher, I want to download project dashboard reports in PDF format, enabling me to conveniently share comprehensive progress updates, milestone achievements, and financial details with my research team, collaborators, and funding bodies.

User Acceptance Test:

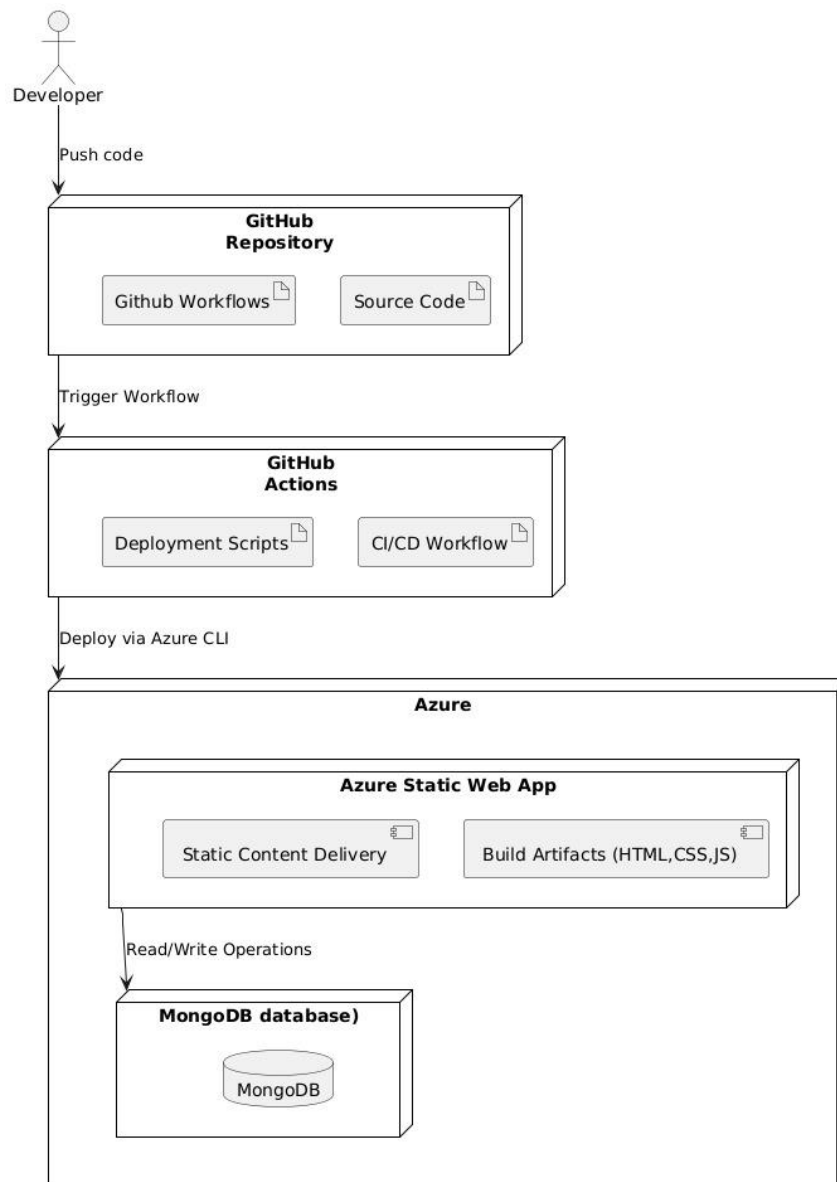
Given that I am on my project dashboard,

When I click the "Download PDF Report" button,

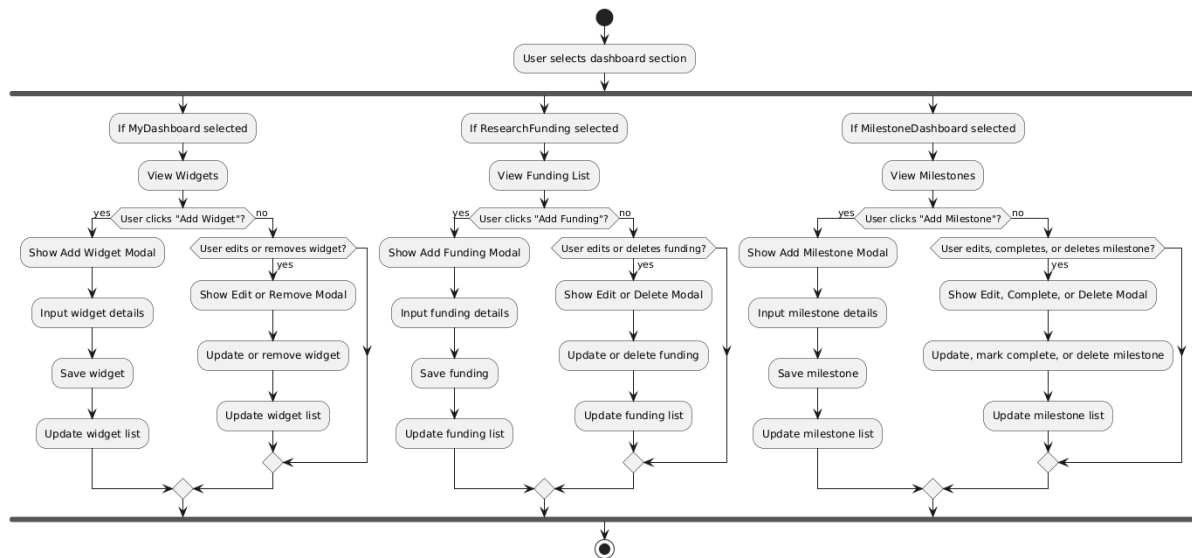
Then a PDF report should be generated and downloaded, containing the latest project progress updates, milestone statuses, and financial summaries.

UML DIAGRAMS

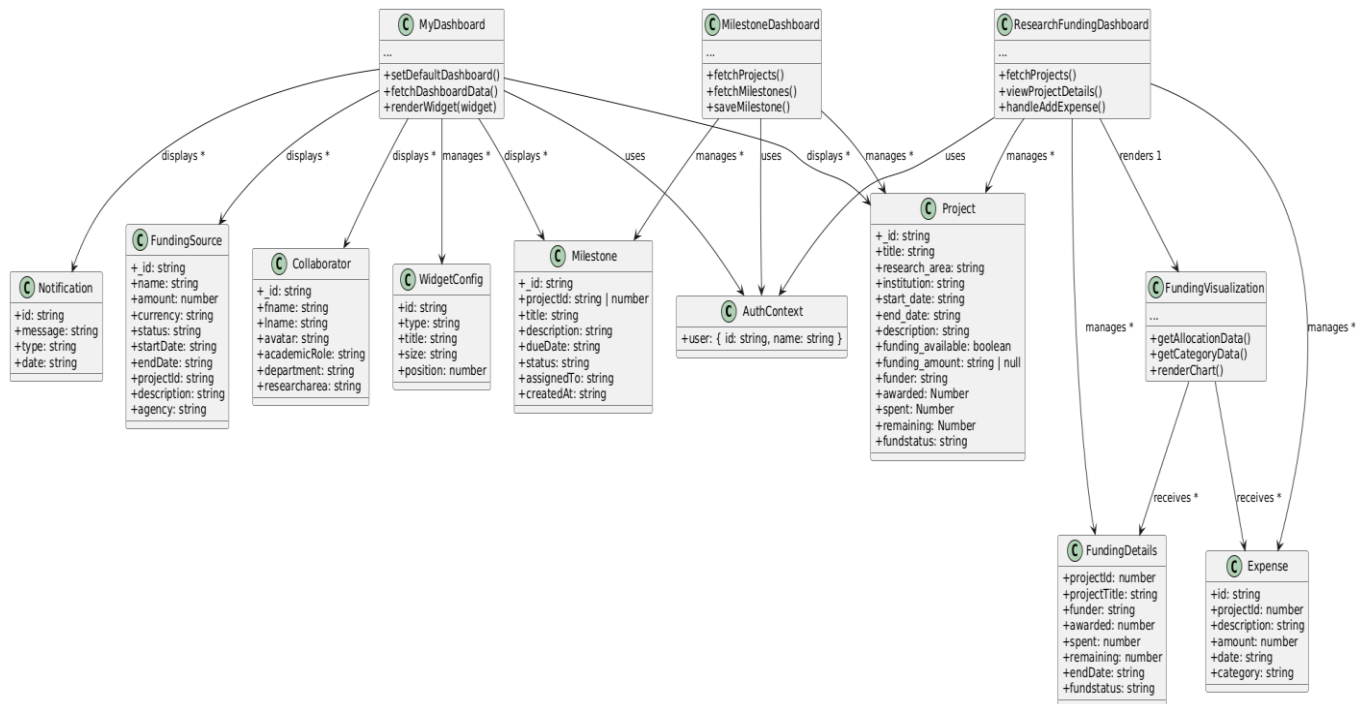
Deployment diagram



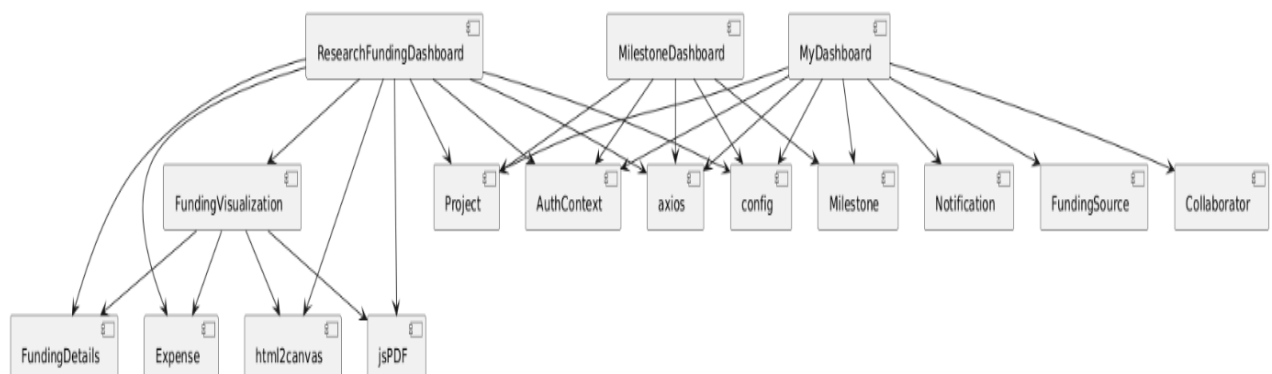
Activity Diagram

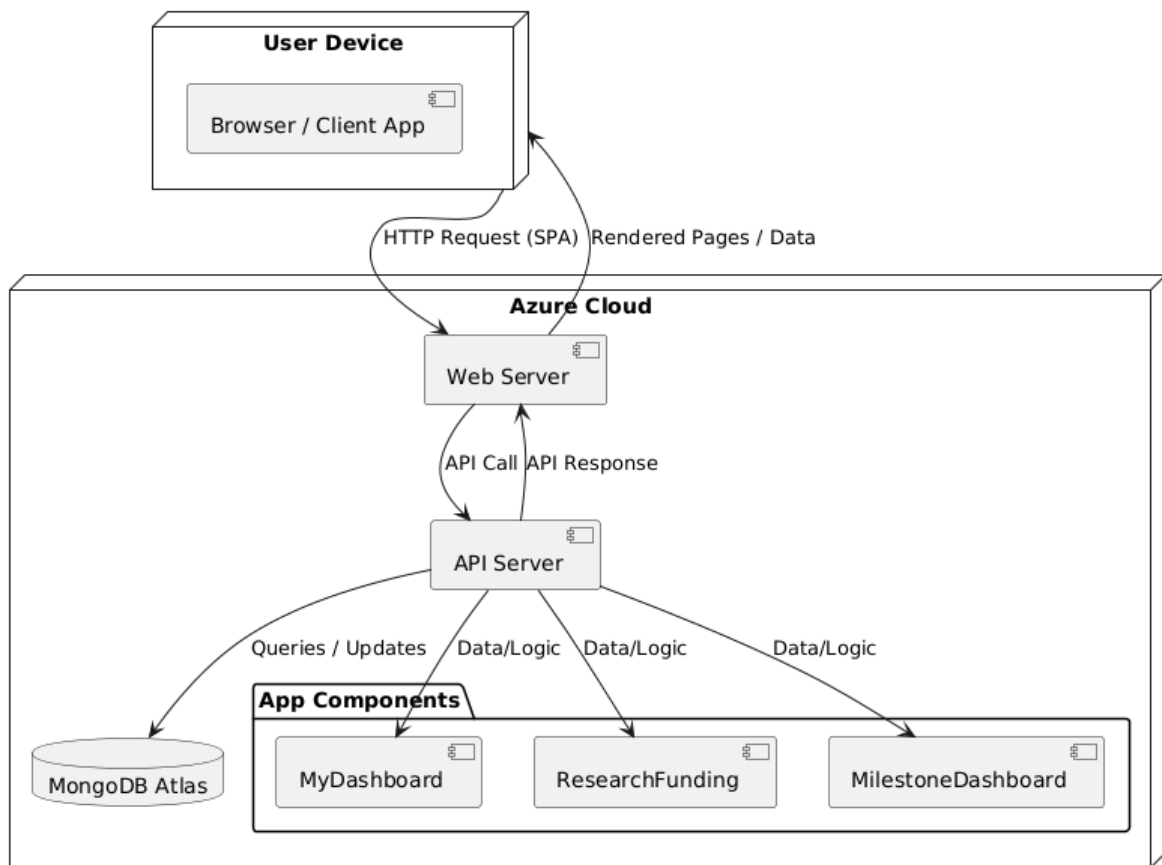


Class Diagram

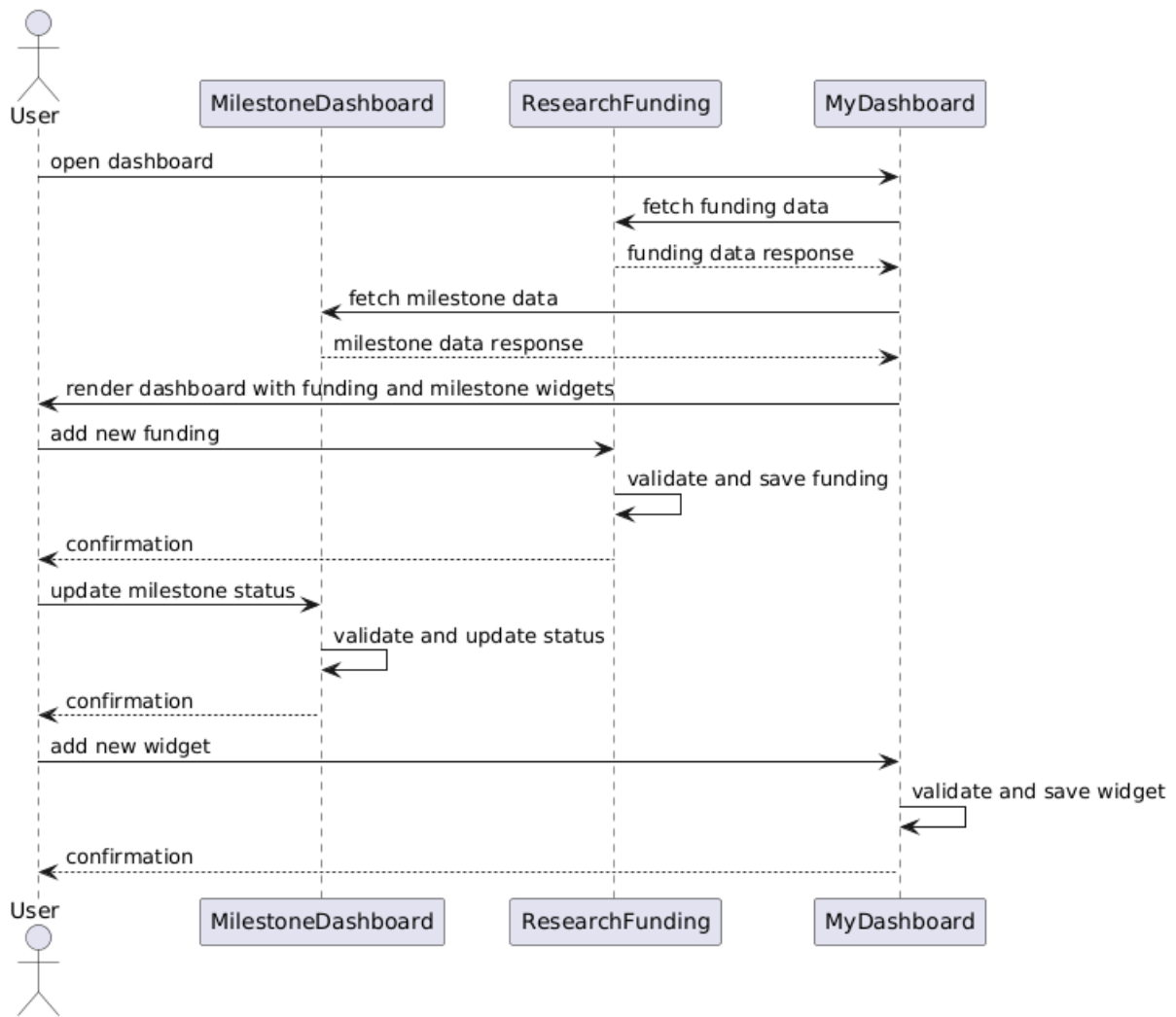


Component Diagram

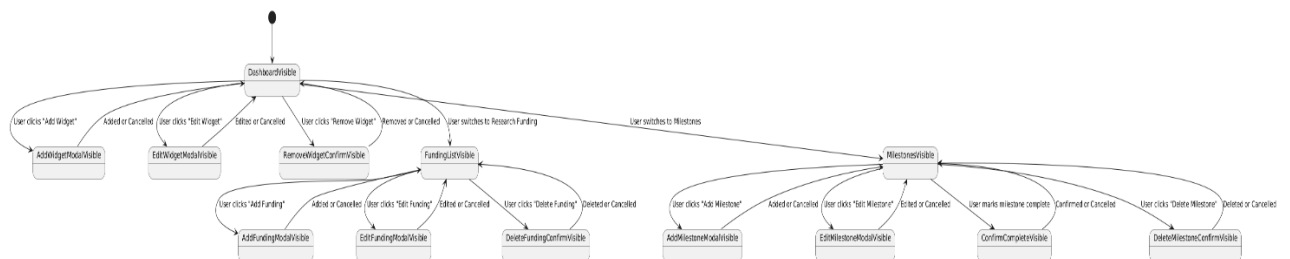




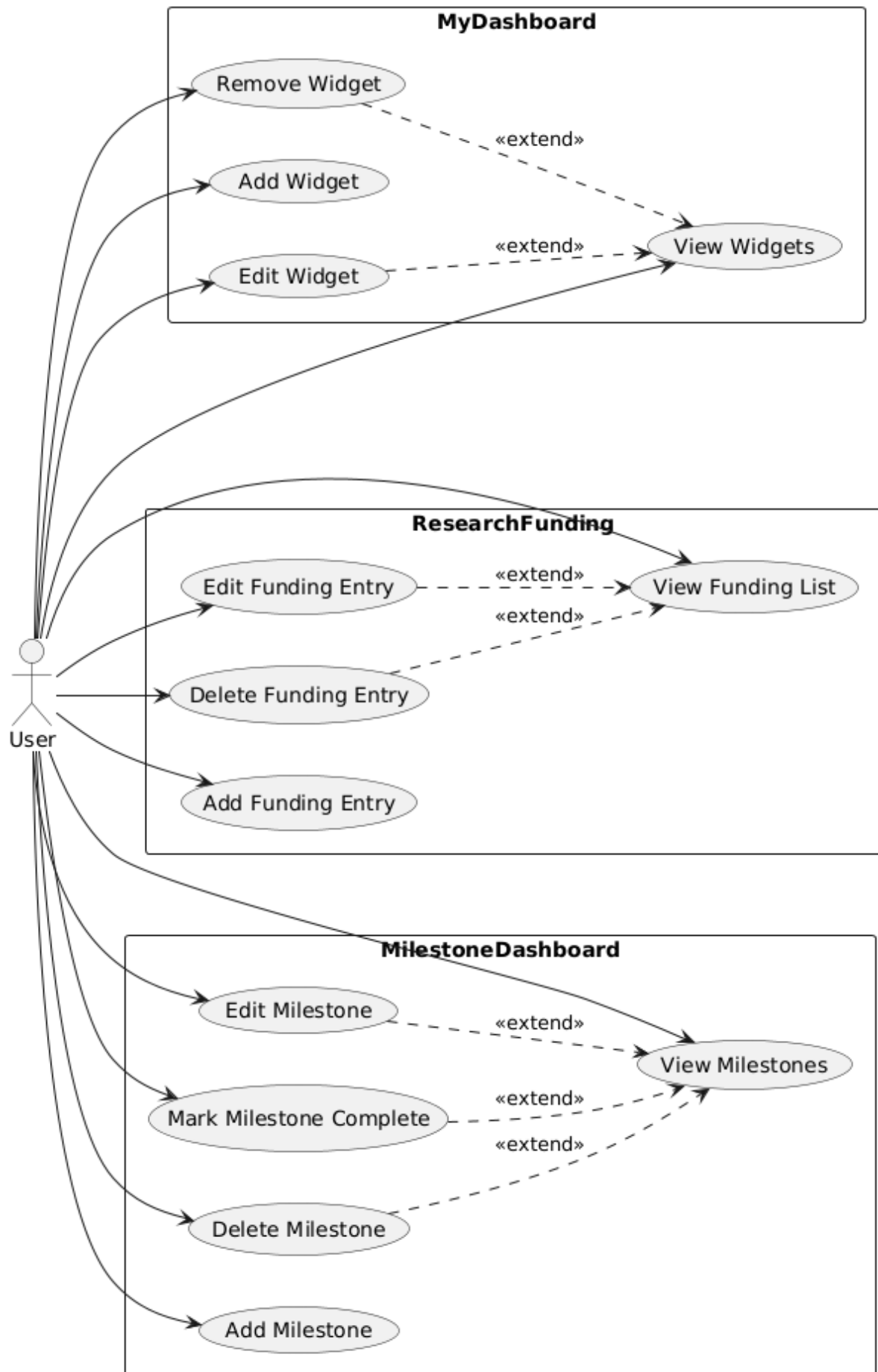
sequence diagram



State Diagram



Use case diagram



MEETINGS

Scrum Meetings

Daily Scrum Meetings

Daily Scrum – 7 May 2025

Facilitator: Renkie Mkhize

Attendees: Renkie Mkhize, Hluma Nziweni, Bongumusa Makhubu, Nhlamulo Mabunda, Tinashe Nganadange, Gabriel Maponya

What was done yesterday:

Created front-end form for project listing creation

Linked form fields to database schema

Initial work on invitation logic

Today's Plan:

Finalize project listing submission

Begin edit functionality on “My Projects”

Design invitation UI flow

Impediments:

Minor issues with date validation on the listing form

Daily Scrum – 8 May 2025

Facilitator: Renkie Mkhize

Attendees: Same as above

What was done yesterday:

Project listing creation completed and pushed to GitHub

Project edit page started

Basic collaborator invitation modal designed

Today's Plan:

Implement invitation search by skill and name

Connect backend for invitation logic

Begin UI for viewing sent invitations

Impediments:

Delay due to confusion in route handling for nested components

Daily Scrum – 9 May 2025

Facilitator: Renkie Mkhize

Attendees: Same as above

What was done yesterday:

Backend for invitations connected

UI for managing invitations started

Some work on admin dashboard layout

Today's Plan:

Complete invitation management logic

Work on delete and edit functionality for projects

Connect admin view to list all users

Impediments:

Issue syncing frontend data state with backend updates

Daily Scrum – 13 May 2025

Facilitator: Renkie Mkhize

Attendees: Same

What was done yesterday:

Researcher can now delete project listings

Admin view of users mostly done

Started on researcher-to-researcher messaging UI

Today's Plan:

Complete messaging functionality

Start work on reviewer dashboard

Add notification logic for new invites

Impediments:

Reviewer routes returning null data

Daily Scrum – 14 May 2025

Facilitator: Renkie Mkhize

Attendees: Same

What was done yesterday:

Messaging flow completed for researcher accounts

Admin dashboard refined

Reviewer dashboard structure created

Today's Plan:

Finalize reviewer dashboard layout

Fix bug preventing reviewer message access

Set up notification pop-ups

Impediments:

Reviewer dashboard not rendering assigned projects properly

Daily Scrum – 15 May 2025

Facilitator: Renkie Mkhize

Attendees: Same

What was done yesterday:

Reviewer dashboard partially fixed

Admin functionality to edit user accounts added

Notification bell component styled and displayed

Today's Plan:

Add reply option to message threads

Debug reviewer project listings

Implement toast notifications

Impediments:

Bug in user role update endpoint (affecting admin edits)

Daily Scrum – 16 May 2025

Facilitator: Renkie Mkhize

Attendees: Same

What was done yesterday:

Notifications are functional and can be dismissed

Admin can now assign reviewers to proposals

Replies in messaging feature functional

Today's Plan:

Complete reviewer evaluation page

Begin code testing coverage

Polish UI for all dashboards

Impediments:

Reviewer still can't view full proposal listing

Sprint Retrospective – 17 May 2025

Facilitator: Renkie Mkhize

Attendees: Full team

Purpose: Reflect on sprint, identify wins, challenges, and improvements

Discussion Points:**What went well:**

Project listing and core researcher functionalities were successfully implemented

Messaging and notifications enhance interactivity

Admin tools now functional and connected to DB

What didn't go well:

Reviewer features were left incomplete

Miscommunication around reviewer dashboard caused delays

Testing and coverage were de-prioritized again

Suggestions for improvement:

Schedule early sprint alignment on complex roles (e.g., reviewer)

Begin writing test cases during implementation

Improve division of front-end and backend tasks

Scrum Planning Meeting – 5 May 2025

Facilitator: Renkie Mkhize

Client: Thisra Pillay (Product Owner)

Attendees: Renkie, Hluma, Bongumusa, Gabriel, Nhlamulo, Tinashe

Purpose: Define sprint scope and user stories provided by the client

User Stories Defined by Client:

Researcher Functionality

Create, edit, delete project listings

Invite collaborators

Manage sent/received invitations

Admin Functionality

Manage all users

View all research proposals

Assign reviewers

View project listings

All Users

Message each other

Receive notifications

Reviewer

View proposals assigned to them

Evaluate submitted projects

Next Steps:

Split user stories into tasks and subtasks

Focus sprint on Researcher + Admin functions first

Reviewer functionality scoped for Sprint 2

Setup initial UI templates and backend endpoints

Sprint Review – 5 May 2025

Facilitator: Thisra Pillay

Attendees: Full team

Completed Work:

Researcher can:

Create, edit, and delete project listings

Invite collaborators

Manage invitations

Respond to collaboration requests

✅ Admin can:

Manage researcher/reviewer accounts

View proposals

Assign reviewers

Access project listings

✅ All users:

Send/receive messages

Get notifications

Reply to messages

Work In Progress:

Reviewer dashboard incomplete

Reviewer cannot view all assigned projects

Reviewer/admin messaging not available

Issues & Observations:

Reviewer dashboard did not list assigned projects

Only collaborators could send messages; admins and reviewers restricted

UI inconsistency in dashboard layout

Feedback Summary:

Positive reception to core features

Encouraged to prioritize reviewer functionality and polish dashboards

Next Steps:

Complete reviewer tools

Enable full message access for all roles

Increase test coverage

Refactor UI for better role-based experience

Backlog Refinement – 16 May 2025

Facilitator: Renkie Mkhize

Attendees: Same team

Reviewed Items:

Broke down large admin tasks into smaller actions (e.g., role update, proposal assignment)

Split reviewer functionality into: view assigned, evaluate proposal, send feedback

Added new task to refactor notification system

Prioritization:

High: Reviewer dashboard, role-based messaging, test coverage

Medium: UI consistency, bug fixing, frontend responsiveness

Clarified User Stories:

Defined acceptance criteria for messaging system

Reviewer stories updated with expected UI flows and database requirements

Re-estimated Tasks:

Reviewer dashboard increased by 2 days due to proposal linking logic

Notification UI re-estimated at 5 hours

New Items Added:

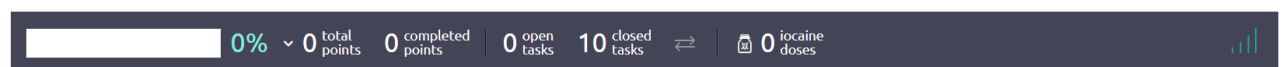
Task for reviewer proposal evaluation submission

Testing and validation of reviewer dashboard

Accessibility enhancements

Sprint Burndown Chart

SPRINT 4 05 May 2025 to 19 May 2025



How this chart works

